

Comparative Analysis of Population Growth and Food Production Challenges: China, Nigeria and Egypt

Bachelor Thesis

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Statutory declaration

I confirm by submitting that this thesis entitled: **Comparative Analysis of Population Growth and Food Production Challenges: China, Nigeria and Egypt** was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references. I agree that the thesis could be made public in accordance with Article 47b of Act No. 111/1998 Coll., on Higher Education Institutions and on Amendments and Supplements to Some Other Acts (the Higher Education Act), and in accordance with the current Directive on publishing of the final theses.

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Abstract

Food security remains an important issue in academic research and public policy. One of its basic dimensions is whether national food availability can be maintained under continuous population growth. This thesis examines the relationship between population growth and national food supply in China, Egypt, and Nigeria through a comparative descriptive analysis.

This study uses secondary national-level data from the World Bank and FAOSTAT and focuses on two indicators: population and food supply (kcal/capita/day). China during 1961–1980 is used as the main historical reference period, post-1980 China as supplementary evidence of later improvement, and Egypt and Nigeria during 2010–2023 as recent comparative cases. The benchmark of 2,100 kcal/capita/day is used only as a simplified descriptive reference point.

The findings show that China during 1961–1980 experienced rapid population growth together with relatively limited and uneven food supply, which remained below the benchmark for most of the period and rose slightly above it only in 1980. After 1980, China showed a different pattern, with slower population growth and clearer long-term improvement in food supply. In Egypt and Nigeria, population continued to grow during 2010–2023 without corresponding improvement in food supply. In Egypt, food supply declined overall although it remained above the benchmark. In Nigeria, population growth was stronger, while food supply also ended lower than at the beginning of the period.

Within the limits of this study, the results suggest that continued population growth combined with weak food supply performance may indicate growing food-related pressure. China is not treated as a direct model for Egypt or Nigeria, but its earlier experience is used as a historical reference for understanding why recent trends in Egypt and Nigeria deserve attention.

Keywords

food security; population growth; food supply; China; Egypt; Nigeria.

Abstrakt

Potravinová bezpečnost zůstává významným tématem akademického výzkumu i veřejné politiky. Jedním z jejích základních rozměrů je otázka, zda lze při dlouhodobém růstu populace udržet dostatečnou dostupnost potravin na národní úrovni. Tato bakalářská práce zkoumá vztah mezi populačním růstem a národní potravinovou nabídkou v Číně, Egyptě a Nigérii prostřednictvím komparativní deskriptivní analýzy.

Práce využívá sekundární data na národní úrovni ze zdrojů World Bank a FAOSTAT a zaměřuje se na dva hlavní ukazatele: populaci a potravinovou nabídku vyjádřenou v kcal na obyvatele za den. Čína v období 1961–1980 je považována za hlavní historické referenční období, vývoj Číny po roce 1980 slouží jako doplňující důkaz pozdějšího zlepšení a Egypt a Nigérie v období 2010–2023 představují současné komparativní případy. Hodnota 2 100 kcal na obyvatele za den je v práci použita pouze jako zjednodušený deskriptivní referenční bod.

Výsledky ukazují, že Čína během let 1961–1980 zažila rychlý populační růst spolu s relativně omezenou a nerovnoměrnou potravinovou nabídkou, která po většinu sledovaného období zůstávala pod uvedenou referenční hodnotou a pouze v roce 1980 ji mírně překročila. Po roce 1980 vykazovala Čína odlišný vývojový vzorec, který se vyznačoval pomalejším populačním růstem a jednoznačným dlouhodobým zlepšením potravinové nabídky. V Egyptě a Nigérii populace v období 2010–2023 dále rostla, aniž by došlo k odpovídajícímu zlepšení potravinové nabídky. V Egyptě potravinová nabídka celkově klesala, i přesto zůstala nad referenční hodnotou. V Nigérii byl populační růst výraznější, přičemž potravinová nabídka byla na konci sledovaného období nižší než na jeho začátku.

V rámci omezení této studie výsledky naznačují, že pokračující populační růst spojený se slabým vývojem potravinové nabídky může poukazovat na rostoucí tlak související s potravinovou situací. Čína není v této práci chápána jako přímý model pro Egypt nebo Nigérii, ale její dřívější zkušenost slouží jako historický referenční případ pro pochopení toho, proč si současné trendy v Egyptě a Nigérii zaslouží pozornost.

Klíčová slova

potravinová bezpečnost; populační růst; potravinová nabídka; Čína; Egypt; Nigérie.

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1 Introduction

Food security has long been an important subject in academic research and public policy. Although this is a broad concept that includes availability, access, stability, and utilization, one of its most basic dimensions is whether a country can maintain sufficient food availability under changing population conditions. Among the many factors that affect food security, population growth is important because it increases overall food needs and can put extra pressure on agricultural output, natural resources, and the wider food system. For this reason, the relationship between population growth and food-related pressure remains a meaningful topic for comparative analysis.

This problem becomes particularly significant when population growth remains high while improvement in national food supply is limited, uneven, or slowing down. A country may not face an immediate food crisis in this situation, but continued population growth can still make food-related problems harder to manage over time. At the same time, food-related pressure should not be judged only by whether the food supply is above or below a fixed reference standard. It is also important to examine how food supply changes over time in relation to continued population growth. From this perspective, trends and direction of change are as important as absolute levels.

Within this broader context, China is an important historical reference case. It experienced serious food-related problems in the late 1950s and early 1960s, followed by rapid population growth and relatively limited food supply from 1961 to 1980. After 1980, however, China showed a different pattern, as food supply improved more clearly and population growth gradually slowed down. This makes the Chinese case useful for descriptive comparison, not because it offers a simple model for other countries, but because it shows how the relationship between population pressure and national food supply can change over time within one country.

By contrast, Egypt and Nigeria are treated in this thesis as recent cases. In recent decades, the two nations have all undergone continuous population growth, and in academic documents they are frequently talked about as having great pressure connected to food, even

though their respective structural conditions are different. In the case of Egypt, the academic literature emphasizes water scarcity, dependence on irrigation, import reliance, and sensitivity to external shocks. In Nigeria, the existing literature places greater emphasis on long-term food poverty, rapid population growth, weak agricultural performance, governance difficulties, conflict, and climate-related pressure. These differences are important because they show that food-related pressure can come from different causes and that the cases examined in this thesis are not the same. Even so, Egypt and Nigeria are still suitable for limited comparison with China's past experience, because they allow this research to examine whether continued population growth has been matched by improvement in national food supply, and whether the observed trends indicate rising pressure over time.

Therefore, this thesis examines the relationship between population growth and national food supply in China, Egypt, and Nigeria through a comparative descriptive method. This research focuses on two macro-level indicators: population and food supply. These indicators cannot capture every aspect of food security, and they also cannot explain all causes of food-related problems. However, they provide a clear and manageable basis for comparing broad trends across different countries and periods. In this thesis, China from 1961 to 1980 acts as the main historical reference time period, China after 1980 is taken as the supplementary proof of subsequent progress, and Egypt and Nigeria from 2010 to 2023 act as the two new comparison cases. In this way, the thesis examines food production challenges indirectly through changes in national food availability under continued population growth.

The aim of this research is not to argue that Egypt and Nigeria are directly repeating China's past, nor to suggest that China provides a universal path for other countries. Instead, the goal is more restricted and descriptive: it is to make a comparison of how population growth and food supply have developed in the three cases, and it is to think about whether current trends in Egypt and Nigeria may show increasing pressure related to food in the future. In this way, this thesis addresses a gap in the existing literature, where these countries are usually studied separately or in different regional and thematic frameworks rather than through one comparative lens.

The rest of this thesis is structured as follows. After the introduction, the thesis presents its aims and research questions. The literature review defines the key concepts, reviews existing research on China, Egypt, and Nigeria, and identifies the research gap. The methodology section explains the research sample, research method, data collection, and data analysis. The results section presents the findings for the three cases. The discussion interprets these results in relation to the existing literature and the limitations of this research, and the conclusion summarizes the main findings within the scope of the thesis.

1.1 Aims of the thesis

This thesis examines the relationship between population growth and national food supply through a comparative descriptive analysis of China, Egypt, and Nigeria. It focuses on changes in population and food supply over time and considers whether continued population growth was followed by improvement, stagnation, or decline in national food availability.

More specifically, these aims are structured around four research questions presented in the methodology section. This thesis has three interrelated objectives. First, it examines China during 1961–1980, which is the main historical reference period, in order to show how rapid population growth coexisted with relatively limited food supply. Second, it takes China after 1980 as the supplementary evidence to display a later time period, in which food supply showed more obvious improvement, and meanwhile population growth had decelerated. Third, this work carries out an examination of Egypt and Nigeria within the period of 2010–2023, in order that it can evaluate whether recent tendencies of population growth and food supply indicate increasing pressure related to food and possible challenges of food production in the future.

This thesis does not have the goal to give explanations for all causes of food insecurity or food production challenges in the chosen countries; it also does not have the goal to put China forward as a direct policy example for Egypt or Nigeria. On the contrary, its range is more restricted: to compare large-scale tendencies in population and food supply

among the three cases and to evaluate what these tendencies may indicate within the frame of a descriptive comparative research.

To achieve these aims, this thesis uses secondary national-level data from the World Bank and FAOSTAT, with particular emphasis on total population and food supply. This method provides a simple and stable basis for comparing population pressure and national food supply among different countries and time periods.

2 Objectives

The main objective of this thesis is to examine the relationship between population growth and national food availability in China, Egypt, and Nigeria through a comparative descriptive analysis. It focuses on whether continued population growth was accompanied by improvement, stagnation, or decline in food supply.

More specifically, this thesis has three objectives. First, it examines China during 1961–1980 as the main historical reference period in order to show how rapid population growth coexisted with relatively limited food availability. Second, it uses post-1980 China as supplementary evidence to show a later period in which food supply improved more clearly while population growth slowed. Third, it examines Egypt and Nigeria during 2010–2023 in order to assess whether recent trends in population growth and food supply suggest growing food-related pressure and possible future food production challenges.

This thesis does not aim to explain all causes of food insecurity or food production challenges in the selected countries, and it does not treat China as a direct model for Egypt or Nigeria. Instead, its objective is more limited: to compare broad trends in population growth and food supply across the three cases and to consider what these trends may suggest within the framework of a descriptive comparative study.

3 Literature Review

3.1 Population growth

Population growth, which is defined as the rise of a nation's population through time, is calculated by the World Bank as the yearly percentage change of the total population (World Bank, 2026). In this thesis, it is regarded as a key demographic factor because rapid population growth can raise food needs and bring pressure to national food systems (World Bank, 2026).

The wider academic literature supports the view that a relationship exists between population growth and the pressure placed on food systems (Villarreal and Stloukal, 2005). Villarreal and Stloukal (2005) argue that a larger population means greater total food needs, and that population dynamics are closely connected with food security and development. Dyson (1994) similarly explains that debates on food production are closely connected to concerns about whether food output can keep pace with population growth. Ehrlich, Ehrlich and Daily (1993) also indicate that the linkage between population growth and food production is one of the core long-term sustainability issues.

Population growth is important in food-related studies not only because it increases the number of people who need to be fed, but also because it may intensify pressure on land, water, infrastructure, and agricultural systems at the same time. When the population grows continuously, governments and food systems may face greater difficulty in maintaining an adequate food supply, especially if agricultural production does not improve at a similar pace. In this sense, population growth is not only a demographic change, but also a factor that may affect broader development conditions.

At the same time, the significance of population growth should not be understood in a purely numerical way. What matters is not only that the population becomes larger, but also how quickly it grows and whether food production and food availability can adjust to that change. A country with strong agricultural growth, technological progress, or ef-

fective food distribution may cope with population growth more successfully than a country with weaker productive capacity. Therefore, in this thesis, population growth is treated as a pressure factor whose effects should be interpreted together with changes in food supply rather than in isolation.

This point is especially relevant for the comparative structure of the present study. China, Egypt, and Nigeria all experienced population growth, but the relationship between that growth and food availability did not develop in the same way across the three cases. For this reason, the concept of population growth in this thesis is used not only as a background demographic indicator, but also as an important analytical starting point for comparing how food-related pressure may emerge under different national conditions.

3.2 Food security

Food security is defined by the FAO in its 1996 World Food Summit framework as existing when all people, at all times, have physical and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life (FAO, 2003).

The academic literature further emphasizes food security as a multidimensional concept, which includes related dimensions such as availability, access, stability, and utilization (Villarreal and Stloukal, 2005). Oluoko-Odingo and Ayiamba (2020) similarly emphasize these four dimensions, stressing that food security must be addressed across all of them rather than only through agricultural production.

In this thesis, although food security includes many dimensions, it is mainly examined through the perspective of national-level food availability, because this dimension corresponds most directly to the data used in this research. In order to support the interpretation of food supply levels, this thesis uses 2,100 kcal/person/day as a simplified international benchmark for minimum energy needs (WFP, 2025). This value is widely used in humanitarian planning and food assistance standards (WFP, 2025). However, the method used by the FAO to measure undernourishment does not rely on a single universal caloric threshold (WHO, 2026; FAO, 2008). Instead, the minimum dietary energy requirement

is calculated according to the demographic structure of the population, including age and sex composition (WHO, 2026; FAO, 2008).

Hence, this thesis uses the 2,100 kcal/person/day value only as a practical and simplified benchmark for descriptive comparison, not as a strict universal threshold in FAO methodology. Because food security is a broad concept, it is important to clarify which part of it is examined in this thesis. In many studies, food security is discussed through a wide range of issues, including household income, prices, food distribution, health conditions, and nutritional outcomes. However, not all of these dimensions can be examined equally in one undergraduate thesis. For this reason, the present study does not attempt to provide a complete assessment of food security in China, Egypt, and Nigeria. Instead, it focuses on national food availability as one important dimension within the broader concept.

This narrower focus does not mean that the other dimensions are unimportant. On the contrary, the literature clearly shows that food insecurity can exist even where aggregate food supply appears sufficient (Villarreal and Stloukal, 2005; Ehrlich, Ehrlich and Daily, 1993). People may still face difficulties because of poverty, unequal distribution, conflict, weak infrastructure, or rising food prices (Villarreal and Stloukal, 2005; Ehrlich, Ehrlich and Daily, 1993). Therefore, when this thesis refers to food-related pressure, it does not mean that national food supply alone fully determines whether a country is food secure or food insecure. Rather, it means that food availability remains one necessary foundation for broader food security.

For the purposes of this thesis, the concept of food security is therefore used in a limited and clearly defined way. It provides the wider theoretical background, while the actual empirical analysis concentrates on national-level food availability measured through food supply data. This distinction is important because it helps maintain consistency between the conceptual discussion and the data used later in the study.

3.3 Food supply (kcal/capita/day)

In the present thesis, food supply is measured using the FAOSTAT Food Balance Sheets “Food supply” indicator, which quantifies food availability in kilocalories per person per

day at the national level (FAO, 2026). In this thesis, this measure is used as a macro-level indicator of food availability rather than actual individual food intake (FAO, 2026). This approach is consistent with the broader academic literature, which treats food availability as one key dimension of food security (Villarreal and Stloukal, 2005). For example, Villarreal and Stloukal (2005) clearly identify it as one of the core pillars.

Nevertheless, national-level food supply measures overlook inequalities in food distribution within countries. This limitation is also consistent with the broader literature, which shows that food insecurity may still exist even when total food availability appears sufficient, especially when access to food is limited by poverty and unequal distribution (Ehrlich, Ehrlich and Daily, 1993; Villarreal and Stloukal, 2005). The use of food supply (kcal/capita/day) in this thesis is especially useful because it provides a simple and internationally comparable indicator for examining long-term trends. Since the study compares different countries and different periods, it is necessary to rely on a measure that is available in a consistent form across all cases. FAOSTAT Food Balance Sheets are suitable for this purpose because they allow broad descriptive comparison at the national level and make it possible to observe whether food availability improved, weakened, or remained unstable over time (FAO, 2026).

At the same time, food supply in kilocalories per person per day should be interpreted carefully. It is an average national indicator, and therefore it does not reflect differences between social groups, regions, or households. A country may show a relatively high food supply level while many individuals still face inadequate diets. In addition, this measure does not directly show dietary quality, diversity, or nutritional balance. Thus, in this thesis, food supply is used as a practical indicator of national food availability, but not as a complete measure of actual food consumption or nutritional well-being.

Even with these limitations, food supply remains useful for the present study because the thesis is concerned with broad macro-level trends rather than household-level outcomes. The main interest here is whether continued population growth was accompanied by stronger, weaker, or stagnant food availability at the national level. In this context, the

food supply indicator provides a manageable and meaningful basis for descriptive comparison across China, Egypt, and Nigeria.

3.4 Relationship between population growth and food security

Although population growth is not the only factor shaping food security, it is generally regarded as a key pressure on food systems (Villarreal and Stloukal, 2005). As Villarreal and Stloukal (2005) point out, a larger population can increase total food demand and make the relationship between food availability and access more important. Dyson (1994) also points out that concerns about food scarcity are closely connected with the balance between population growth and food production, especially in the context of per capita food production. Ehrlich, Ehrlich, and Daily (1993) argue that whether food output can keep pace with population growth remains one of the core questions in the discussion of sustainable development. In the African context, Falkenmark (1990) further argues that rapid population growth increases pressure on already limited water resources, thereby raising the risk of an unstable domestic food supply.

These viewpoints form the analytical foundation of this thesis, which treats population growth not as the only cause of food insecurity, but as a key pressure factor interacting with agricultural production, natural resources, and institutional capacity (Villarreal and Stloukal, 2005; Dyson, 1994; Ehrlich, Ehrlich and Daily, 1993; Falkenmark, 1990). The relationship between population growth and food security is complex and should not be reduced to a simple one-directional explanation. Population growth does not automatically cause food insecurity, and food insecurity is never the result of only one variable. Nevertheless, the literature repeatedly suggests that rapid population growth can make food-related challenges more difficult to manage, especially in situations where agricultural production, natural resources, or institutional capacity are already under pressure (Villarreal and Stloukal, 2005; Dyson, 1994; Ehrlich, Ehrlich and Daily, 1993; Falkenmark, 1990). In such cases, demographic growth may increase the scale of existing problems and reduce the margin available for adjustment (Villarreal and Stloukal, 2005; Dyson, 1994; Ehrlich, Ehrlich and Daily, 1993; Falkenmark, 1990).

This relationship is particularly important when food supply does not improve at the same pace as population. If the number of people grows quickly while food availability grows slowly, remains unstable, or even declines, pressure on the food system may become more visible over time. For this reason, the present thesis does not treat population growth as an isolated explanatory factor. Instead, it studies population growth together with food supply in order to see whether the overall direction of change suggests rising or easing pressure on national food availability.

This approach is also useful for comparison across cases. The literature shows that the causes of food-related difficulty may differ greatly from country to country, but the co-existence of rapid population growth and weak food supply performance can still serve as an important warning sign (Villarreal and Stloukal, 2005; Dyson, 1994; Ehrlich, Ehrlich and Daily, 1993; Falkenmark, 1990). Therefore, in this thesis, the relationship between population growth and food security is approached through the narrower lens of population pressure and national food availability, while still recognizing the wider structural context discussed in the literature.

3.5 Operationalization of concepts in this thesis

According to these definitions, this thesis uses population growth as an indicator of demographic pressure and food supply as a simplified measure of national food availability. Its goal is not to evaluate every aspect of food security, but rather to examine how changes in population growth are connected with changes in national food supply in China, Egypt, and Nigeria. This method is consistent with the existing literature, which treats food security as a multidimensional concept and emphasizes food availability as a basic pillar for understanding the interaction between population growth and food production challenges (FAO, 2003; Villarreal and Stloukal, 2005). Hence, this research uses official FAOSTAT and World Bank indicators as its main empirical basis (World Bank, 2026; FAO, 2026). At the same time, it draws support from the broader academic literature to explain how population growth may contribute to greater food security pressure over time (Dyson, 1994; Ehrlich, Ehrlich and Daily, 1993).

Operationalizing the concepts in a clear way is important for keeping the study focused and methodologically consistent. Since food security is a multidimensional concept, the thesis must define which dimensions can realistically be examined with the available data. For that reason, this study translates the broader concept into two measurable indicators: population growth and food supply. Population growth represents demographic pressure, while food supply represents national food availability. These operational choices make it possible to move from general theoretical discussion to a concrete empirical comparison.

This operationalization also helps maintain a clear boundary for the thesis. Rather than trying to measure every possible factor related to food insecurity, the study adopts a simpler framework that can be applied consistently across China, Egypt, and Nigeria. This is especially important because the thesis compares different countries and historical periods. A narrow but consistent framework is more suitable for such comparison than a wider framework that cannot be measured equally across all cases.

At the same time, this operationalization does not deny the importance of the other dimensions of food security. Instead, it reflects a deliberate research choice. The thesis acknowledges that access, distribution, governance, conflict, and environmental constraints matter greatly, but it examines them mainly as background factors drawn from the literature review. In this way, the study remains modest in scope while still connecting the empirical analysis to the wider food security debate.

3.6 Current work in China

Existing research on China generally agrees that the country faced a serious food crisis in the late 1950s and early 1960s, which was closely connected with agricultural failures and institutional weaknesses (Yang, 2008). Yang (2008) points out that the collapse of grain output was central to this crisis, and that declining grain availability—together with a distribution system that favored urban areas—directly contributed to widespread famine. Yang (2008) emphasizes that no single factor can fully explain the famine. Rather,

failures in central planning, excessive grain procurement, labour shifts away from agriculture, and adverse weather all interacted during the Great Leap Forward (Yang, 2008). This balanced interpretation is especially valuable for the present thesis because it shows that China's food shortage in this period did not result from one single cause, but from the interaction of production, policy, and distribution problems (Yang, 2008).

Cheng (1973) provides another perspective by describing agriculture as the "Achilles' heel" of China's economy at the beginning of the 1970s. Cheng (1973) notes that although industrial output had increased much more quickly, agricultural production between 1952 and 1972 grew at only about 2 percent per year, which meant that staple food supply remained highly sensitive to weather and harvest fluctuations. Furthermore, Cheng (1973) notes that as China's population approached 900 million, repeated harvest failures could create serious economic and political difficulties. His article therefore helps explain why food production remained a major concern even after the most severe famine years had passed (Cheng, 1973).

Banister (1984) examines population policy and demographic trends in China more directly. She shows that the country experienced rapid population growth from the 1950s through much of the 1960s, with annual growth rates of around 2 percent or higher in non-famine years (Banister, 1984). By the early 1970s, Chinese leaders had become convinced that this expansion was placing pressure on living standards, because per capita supplies of basic goods—especially food—had not improved substantially compared with earlier decades (Banister, 1984). This analysis is especially relevant to this thesis because it directly links demographic pressure to per capita food availability and helps explain why population policy became increasingly important by the end of the 1970s (Banister, 1984). At the same time, Banister (1984) does not attribute China's food problems entirely to population growth. On the contrary, she stresses that both increased production and reduced population growth were necessary for improving living standards (Banister, 1984).

This interpretation is also consistent with China's official statements on population policy, which identify population growth, limited farmland, and pressure on economic and

social development as key background factors behind family planning (Ministry of Foreign Affairs of the People's Republic of China, 2002). These materials note that China experienced a second major birth peak from 1962 to 1972, and that the contradiction between a large population, limited cultivated land, and development needs became more obvious over time (Ministry of Foreign Affairs of the People's Republic of China, 2002). In this thesis, population policy is not treated as a main subject of political evaluation, but rather as historical background that helps explain why population growth became an urgent issue in China by the end of the 1970s (Ministry of Foreign Affairs of the People's Republic of China, 2002).

Later research shifted attention from crisis to transition. Ghose (2014), for example, treats the founding of the People's Republic in 1949 and the economic reforms of the late 1970s and 1980s as two major turning points in China's agricultural history. He argues that the introduction of the Household Responsibility System brought major improvements in agricultural output and productivity, making it possible for China to produce most of its food requirements (Ghose, 2014). However, Ghose (2014) also notes that rapid population growth and dietary change increased food demand, thus creating new pressures on land, water resources, and long-term self-sufficiency. This perspective is especially relevant to the present thesis because it shows that improvements in China's food security did not remove demographic pressure; rather, the nature of these challenges changed over time (Ghose, 2014).

Dong et al. (2024) provide a more recent policy-oriented overview of China's food security experience. They stress that food security remains a national priority, shaped by both historical memory and continuing structural limits—such as feeding nearly 20 percent of the world's population with less than 10 percent of global arable land and about 6 percent of global freshwater resources (Dong et al., 2024). The report further links post-1978 reforms to major growth in agricultural production and later reductions in undernourishment, especially through institutional reform, incentive changes, and technological progress (Dong et al., 2024). In this way, it complements Ghose's long-term analysis by tracing the development of China's food security challenges from famine prevention and

basic grain provision to broader issues of sustainability, self-sufficiency, and stable supply under changing domestic and global conditions (Dong et al., 2024).

Taken together, these studies show a clear pattern in the Chinese case. The literature strongly supports the view that China experienced a severe food crisis in the late 1950s and early 1960s, caused by agricultural collapse, institutional problems, and unequal food distribution (Yang, 2008). Several studies also suggest that rapid population growth increased pressure on per capita food availability and made the food problem more difficult to manage (Banister, 1984; Cheng, 1973; Ghose, 2014). Moreover, China's later improvement in food security resulted from a combination of agricultural reform, productivity growth, and demographic change, rather than from any single measure (Ghose, 2014; Dong et al., 2024). These findings are directly relevant to the present thesis because they provide the historical and analytical basis for comparing China's earlier experience with recent food production challenges in Egypt and Nigeria.

An important strength of the literature on China is that it does not treat the country's food history as a simple story of shortage followed by success. Instead, the studies reviewed above show a more complex picture in which crisis, demographic pressure, institutional change, and later agricultural improvement are all connected (Yang, 2008; Banister, 1984; Ghose, 2014; Dong et al., 2024). This complexity is valuable for the present thesis because it supports a two-stage understanding of the Chinese case. On the one hand, the literature explains why food-related pressure was severe in the period surrounding the late 1950s, 1960s, and early 1970s (Yang, 2008; Cheng, 1973; Banister, 1984). On the other hand, it also helps explain why later developments should be understood as a different stage rather than merely as a continuation of the earlier one (Ghose, 2014; Dong et al., 2024).

Another important point in the Chinese literature is that improvement in food security is not presented as the outcome of a single policy alone. The reviewed studies suggest that later progress emerged from the interaction of agricultural reform, institutional incentives, productivity growth, and changing demographic conditions (Ghose, 2014; Dong et al., 2024). This is relevant to the present thesis because it supports a cautious interpretation

of the Chinese experience. China's later improvement is analytically important, but it should not be reduced to a simple lesson that can be mechanically applied elsewhere.

For this thesis, the Chinese literature is useful not because it offers a perfect policy model, but because it provides a historically grounded case in which the relationship between population pressure and food availability changed substantially over time (Yang, 2008; Banister, 1984; Ghose, 2014; Dong et al., 2024). This makes China especially valuable as a reference case. It allows the thesis to compare not only different countries, but also different stages within one country, which strengthens the overall comparative logic of the study.

3.7 Current work in Egypt

Existing research on Egypt holds that this nation's food security challenge is not most appropriately comprehended as a simple total lack of food, but instead as a structural problem that is formed by natural-resource restrictions, reliance on irrigation, and, in more recently published works, strong susceptibility to external market impacts (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). Stokes, Scozzaro and Haller (2010) hold the view that Egypt is a typical example of physical water shortage, in which the naturally low water occurrence has been counterbalanced by large irrigation and water handling to keep up agricultural output. They additionally point out that this method has already allowed short-term reduction of food shortages, although climate change and population increase may strengthen burdens on the coming food supply (Stokes, Scozzaro and Haller, 2010). The same study emphasizes that Egypt has a very large dependence on irrigation from the Nile River; this irrigation maintains crop production in the situation of yearly rainfall being extremely low, therefore making this country one of the nations with the most irrigation areas in the world (Stokes, Scozzaro and Haller, 2010). This method has decreased sensitivity to environmental uncertain factors such as drought (Stokes, Scozzaro and Haller, 2010). However, the authors remind that water use already gets close to 92 percent of renewable boundaries, thus limiting the further enlargement of the existing model (Stokes, Scozzaro and Haller, 2010). Therefore, Egypt

keeps food supply through high-intensity irrigation under increasingly strict natural-resource restrictions (Stokes, Scozzaro and Haller, 2010).

El-Khoury (2013) gives a wider statistical view by inspecting agriculture and food security in Arab countries through the use of selected indicators. In this regional contrast, Egypt stands out because of its large population size, large agricultural labour force, and comparatively important agricultural sector (El-Khoury, 2013). This research indicates that the self-sufficiency rate of cereals in Egypt was 56.78 percent in 2010, which only satisfied part of the country's demand, and that the food import amount exceeded 8.5 billion US dollars in that year (El-Khoury, 2013). This result highlights the country's very heavy dependence on outside markets (El-Khoury, 2013). These indicators show that Egypt's food security problem cannot be reduced to extensive overall inadequate nutrition only, and is also connected with structural reliance on imports and restricted self-sufficiency in key food goods (El-Khoury, 2013).

Abay et al. (2022) make this distinction even more obvious in their IFPRI report. They argue that Egypt, on account of its high dependence on food imports, was especially vulnerable to the global crisis caused by the Ukraine war (Abay et al., 2022). As the world's biggest wheat-importing country, Egypt satisfies around 50 percent of its own wheat needs by means of imports, and wheat is a key staple grain crop which accounts for 39 percent of every person's calorie intake (Abay et al., 2022). This great dependence makes this nation particularly responsive to changes in world wheat prices and supply chains (Abay et al., 2022). Abay et al. (2022) further argue that although the present worldwide crisis did not produce the most remarkable influence on Egypt's whole GDP, it exerted direct influence on the agrifood system. Their analysis reveals GDP and employment losses in this sector, and poorer and rural households are particularly vulnerable to reduced consumption (Abay et al., 2022). They therefore draw the conclusion that, although the cost of a healthy diet did not experience a large increase, decreasing real incomes widened the distance between real household consumption and the needs of a healthy diet (Abay et al., 2022). This viewpoint is important for the current thesis, because it demonstrates that food security pressures in Egypt may become stronger through income, price,

and import channels, even when the problem is not explained simply as a problem of total food lack (Abay et al., 2022).

Put together, these studies show that Egypt's food security challenges come from a combination of resource restrictions, partial dependence on domestic production, and vulnerability to import shocks—not merely a simple total food lack at the current time (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). Although Egypt has maintained food supply through irrigation, import capacity, and policy support systems, the research literature shows that there are lasting pressures coming from population increase, water scarcity, and global market instability (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). This view is consistent with the current thesis, which studies whether quick population growth and tense food supply tendencies may show coming food security risks, without putting forward a current serious crisis (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022).

A key point that emerges from the literature on Egypt is that the country's food situation should be understood less as a simple problem of national caloric shortage and more as a structurally constrained system under pressure (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). The studies reviewed above suggest that Egypt has maintained food availability through a combination of irrigation, imports, and state capacity, but this does not remove underlying vulnerabilities (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). On the contrary, it means that food security depends heavily on systems that may become harder to sustain if resource pressure, external shocks, or demographic growth continue (Stokes, Scozzaro and Haller, 2010; Abay et al., 2022).

This is important for the present thesis because Egypt's case differs clearly from China's earlier experience. In China's historical case, the literature emphasizes low food availability and severe internal crisis (Yang, 2008; Cheng, 1973; Banister, 1984). In Egypt, by contrast, the literature points to a situation in which food supply can remain relatively high in aggregate terms while still being exposed to structural risk (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). Therefore, Egypt is relevant for the

thesis not because it reproduces China's past, but because it represents a different form of pressure in which food availability may weaken even without falling into an immediately critical caloric level.

The literature also helps explain why recent trends in Egypt deserve attention. If a country already faces water scarcity, import dependence, and sensitivity to global shocks, then continued population growth may increase pressure on an already constrained system (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). In this sense, the Egyptian case supports the broader argument of this thesis that food-related pressure should not be judged only by extremely low food supply, but also by whether food availability is improving or weakening under continued demographic expansion (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022).

3.8 Current work in Nigeria

Currently existing research about Nigeria shows that food insecurity is a long-standing, grave problem of society and the economy (Ozughalu and Ogwumike, 2015). Ozughalu and Ogwumike (2015) put forward the argument that, although the nation has plentiful human and material resources, it still continues to struggle with insufficient food intake, wide-ranging poverty, and long-lasting lack of food. They report that hunger, starvation, and malnutrition are widespread in the whole country, and that food poverty differs greatly among regions (Ozughalu and Ogwumike, 2015). Their analysis also discovers regional distinctions in food poverty lines, together with a high occurrence at the national level (Ozughalu and Ogwumike, 2015). It points out that the problems that do not stop existing in the gaining of food that is nutritious remain serious challenges (Ozughalu and Ogwumike, 2015). Therefore, they put forward policies that focus on strengthening food obtainability and lowering the occurrence rate of food poverty (Ozughalu and Ogwumike, 2015).

One other branch of research investigates population-related pressures in a more direct way. Nwoke (2016) demonstrates that the population of Nigeria has grown fast in past decades, with yearly growth rates increasing from 2.8 percent during 1960–1965 to 3.1

percent during 1970–1975 and remaining above 3 percent after that (Nwoke, 2016). The expansion of population still kept going strongly after 2006, and this highlighted the continuous pressure that fast growth puts on the food system (Nwoke, 2016). Nwoke (2016) furthermore emphasizes accelerated rural-to-urban population movement and fast urban development since independence: the proportion of the urban population climbed from only 11 percent in 1952 to 28 percent by 1985, making it one of the world's fastest urbanization processes during the 1970s (Nwoke, 2016). These understandings are relevant to the current thesis because they show that population pressure in Nigeria is formed not only by fast population growth, but also by long-run urbanization and migration trends (Nwoke, 2016).

Howard, Simmons, and Flowers (2019) provide a newer policy-related view. They describe Nigeria as a country that faces obvious instability and danger but still possesses quite a big economic potential (Howard, Simmons and Flowers, 2019). They point out that agricultural growth speeds have not caught up with increasing food needs, therefore leaving countryside poverty high and food insecurity persistent, while reliance on food imports has become larger (Howard, Simmons and Flowers, 2019). This analysis has a special connection to the current thesis because it shows how Nigeria's food security pressures are formed through the mutual action of increasing food need, restricted agricultural development, and feeble structural conditions (Howard, Simmons and Flowers, 2019). The report moreover attributes the current food security issue to a number of factors that strengthen each other (Howard, Simmons and Flowers, 2019). Insufficient public spending on agriculture, frequent policy changes, weak governance, and too little investment in infrastructure have all limited the nation's capability to transform its food system (Howard, Simmons and Flowers, 2019). Economic shocks—especially the steep oil price fall in 2014/15—made these difficulties more serious by cutting import volumes, government expenditures, and the supply of food and agricultural products (Howard, Simmons and Flowers, 2019). Besides this, Howard, Simmons, and Flowers (2019) argue that climate change and conflict have already emerged as key components of Nigeria's food security problems. Drought, flood, and changed seasonal patterns destroy farming output and household income, while continuing conflicts and displacement in many areas stop

people from getting productive resources and market access (Howard, Simmons and Flowers, 2019). In addition, the report points out that Nigeria's fast-growing population has raised demand not only for basic foods but also for meat and milk products, therefore increasing pressure upon land and water resources in some regions (Howard, Simmons and Flowers, 2019).

Putting all together, these researchers identify a steady pattern in Nigeria's food security challenges. Existing studies point out that severe food poverty and insecurity have existed for a very long time (Ozughalu and Ogwumike, 2015). They also record many decades of rapid population growth, together with faster urbanization and rural-to-urban population movement, which have made pressure on the food system stronger (Nwoke, 2016). More recent analyses show how weak agricultural growth, market pressures, imports, governance defects, climate stress, and conflict interact with population pressure to form current challenges (Howard, Simmons and Flowers, 2019).

These research results have a direct connection to the current thesis; they show that Nigeria's food security challenges go further than food poverty to wider pressures that are formed by rapid population growth, urbanization, weak agricultural performance, and structural weaknesses (Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019). The literature on Nigeria presents a case in which food-related pressure appears more persistent and multidimensional (Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019). Unlike the Egyptian case, where aggregate food availability remains relatively high despite structural constraints, the Nigerian literature more directly emphasizes long-term food poverty, rapid demographic growth, and weak agricultural performance (Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019). This makes Nigeria especially important for the present thesis, because it shows how food-related pressure may deepen when population increase, structural weakness, and insufficient agricultural improvement occur together (Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019).

Another useful feature of the Nigerian literature is that it links demographic change with wider social transformation (Nwoke, 2016; Howard, Simmons and Flowers, 2019). The reviewed studies suggest that rapid population growth is not occurring in isolation, but alongside urbanization, migration, governance challenges, climate stress, and conflict (Nwoke, 2016; Howard, Simmons and Flowers, 2019). This strengthens the relevance of Nigeria for the present study. It shows that the pressure associated with population growth is likely to interact with several other structural difficulties, making food-related problems harder to resolve over time (Nwoke, 2016; Howard, Simmons and Flowers, 2019).

For the purpose of this thesis, Nigeria is therefore not treated simply as a country with rapid population growth, but as a case where demographic expansion occurs under broader structural fragility (Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019). This helps explain why the Nigerian case is analytically useful in comparison with China. It does not mirror China's history directly, but it offers a contemporary example of how continued population growth and weak food system performance may coincide in a way that deserves careful attention (Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019).

3.9 Research gaps

Current research gives useful understandings on the interaction between food security, food production, and population pressures, but it still leaves a number of important gaps that are directly related to this current thesis. Although the wider academic circle accepts that food security has multiple dimensions, in which population increase raises the demand for food and brings pressure to systems, especially when production, access, and natural resources cannot keep the same speed (Villarreal and Stloukal, 2005; Ehrlich, Ehrlich and Daily, 1993; Dyson, 1994), these pieces of research still mostly remain theoretical in character or take the whole world as their research scope (Villarreal and Stloukal, 2005; Ehrlich, Ehrlich and Daily, 1993; Dyson, 1994). Therefore, they are not able to give a concentrated comparative framework for studying the interaction between population growth and food supply in the specific instances of China, Egypt, and Nigeria.

In the case of China, the academic documents are comparatively abundant but frequently scattered among different research directions. Several studies investigate the 1959–1961 great hunger, putting emphasis on grain production decline, planning mistakes, and distributional bias (Yang, 2008). Other scholars discuss the weaknesses of agriculture in the early reform period, population policy, or the long-term improvements in food security after the Household Responsibility System (Cheng, 1973; Banister, 1984; Ghose, 2014; Dong et al., 2024). Even though these studies have their value, they seldom place China's historical transition into a direct comparative framework together with current African cases facing population growth and food system pressures (Yang, 2008; Cheng, 1973; Banister, 1984; Ghose, 2014; Dong et al., 2024).

As for Egypt, the existing academic documents tend to emphasize structural restrictive factors such as water shortage, reliance on irrigation, dependence on imports, and vulnerability to global price shocks (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). These studies are helpful in showing that Egypt's food security challenges go further than simple aggregate caloric shortage (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022). However, they do not directly compare Egypt's current population pressure and food supply trends with China's past historical transition from food insecurity (Stokes, Scozzaro and Haller, 2010; El-Khoury, 2013; Abay et al., 2022).

Regarding Nigeria, the academic literature points to long-term food poverty, rapid population growth, urbanization, and more recent structural pressures connected to weak agricultural output, imports, governance defects, climate stress, and conflict (Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019). But these studies still mostly do not connect Nigeria to the China case, and they do not carry out comparative research on Nigeria with a historical example of a country that once confronted very serious food insufficiency under rapid population growth but later improved its food supply (Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019). Another important research gap is that the majority of current academic documents treat China, Egypt, and Nigeria either as individual national cases or as components of wider regional discussions. For example, studies on Egypt are frequently placed in the framework of Arab food security or global import-shock vulnerability, while studies on

Nigeria usually emphasize poverty, demographic change, or current instability (El-Khoury, 2013; Abay et al., 2022; Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019). Therefore, there is limited research that directly compares these three countries through a consistent analytical lens focused on population growth and food production or availability challenges (El-Khoury, 2013; Abay et al., 2022; Ozughalu and Ogwumike, 2015; Nwoke, 2016; Howard, Simmons and Flowers, 2019).

One more gap relates to the connection between past experiences and present policy learning. The research documents about China describe how famine, population pressure, agricultural reforms, and subsequent productivity growth shaped the long path of the country's food security development (Yang, 2008; Banister, 1984; Ghose, 2014; Dong et al., 2024). At the same time, research on Afro-China cooperation argues that food security is still an important but not fully used area of China–Africa cooperative partnerships, and that opportunities exist in research, technology transfer, agricultural processing, and wider development cooperation (Oluoko-Odingo and Ayiemba, 2020). However, these two strands of literature have seldom been combined to discuss what specific experiences from China's historical development may be useful for countries such as Egypt and Nigeria under present population pressure (Yang, 2008; Banister, 1984; Ghose, 2014; Dong et al., 2024; Oluoko-Odingo and Ayiemba, 2020).

This current thesis deals with these gaps by means of a small and focused comparative analysis. Instead of including all political, institutional, or nutritional aspects of food safety, this research studies China, Egypt, and Nigeria through two core indicators: population growth and national-level food supply. This method connects China's experience of a limited food supply when the population increased fast to the current pressure on supply in Egypt and Nigeria, but it does not say these examples are the same. Therefore, this thesis gives a clear-cut framework that links population pressures, national food supply, and production challenges among these three different but analytically connected nations.

Another research gap relates to the empirical emphasis that existing studies have. A large part of the existing research examines food security through a wide collection of political,

institutional, environmental, and economic elements, often in very great detail. Although these factors are important, they are seldom analyzed in a simple, comparable framework that works across different countries (Villarreal and Stloukal, 2005; Ehrlich, Ehrlich and Daily, 1993; Stokes, Scozzaro and Haller, 2010; Howard, Simmons and Flowers, 2019). Furthermore, a large number of studies place emphasis on contexts that are specific to each country or on thematic analyses in depth, which creates a hindrance to direct comparison (Yang, 2008; El-Khoury, 2013; Ozughalu and Ogwumike, 2015). By comparison, this thesis uses secondary macro-level data and a narrower comparative angle, which focuses on the relation between population growth and national food supply. Other elements identified in existing research, such as water shortage, dependence on imports, governance, conflict, and policy change, are recognized as important background conditions but are not studied in this paper with the same empirical depth (Stokes, Scozzaro and Haller, 2010; Abay et al., 2022; Howard, Simmons and Flowers, 2019).

The research gaps identified above show that the existing literature is rich in country-specific detail but less developed in terms of direct comparison across the three selected cases (Yang, 2008; Stokes, Scozzaro and Haller, 2010; Ozughalu and Ogwumike, 2015). This is especially important for the present thesis because its purpose is not to replace detailed national studies, but to connect them through a simpler comparative framework. In other words, the contribution of this thesis lies less in discovering a completely new issue and more in bringing together several strands of literature that are usually discussed separately (Yang, 2008; El-Khoury, 2013; Nwoke, 2016). A second important gap concerns the level of analysis. Many studies examine food security at a broad conceptual level or through highly specific national factors, but fewer studies compare how demographic pressure and national food availability move together over time in a straightforward descriptive way (Villarreal and Stloukal, 2005; Ehrlich, Ehrlich and Daily, 1993; Dyson, 1994). This is where the present thesis positions itself. By focusing on macro-level indicators, it does not try to compete with more detailed political, institutional, or nutritional studies. Instead, it provides a narrower perspective that can make cross-case comparison more manageable.

Finally, the research gap also lies in the limited use of historical reference in relation to contemporary cases. The literature on China is often discussed in its own historical and policy context, while the literature on Egypt and Nigeria is usually framed in present-day structural terms (Yang, 2008; Ghose, 2014; Stokes, Scozzaro and Haller, 2010; Howard, Simmons and Flowers, 2019). These bodies of work are rarely brought into one analytical discussion. The present thesis addresses this gap in a modest way by using China as a historical reference point and Egypt and Nigeria as recent comparison cases. In doing so, it does not claim equivalence between the cases, but it does suggest that comparison across time and context can still be useful for identifying patterns of food-related pressure (Yang, 2008; Stokes, Scozzaro and Haller, 2010; Howard, Simmons and Flowers, 2019).

4 Methodology

4.1 Research sample

This thesis examines three countries—China, Nigeria, and Egypt—which were selected because of their relevance to the challenges of population growth and food production. China serves as a historical reference case, while Nigeria and Egypt provide contemporary cases for comparison.

China was chosen because it provides an important historical case for studying the relationship between population growth and food-related pressure. Although the grave 1959–1961 famine is discussed as historical background in this thesis, the main period of analysis for China covers 1961–1980. During this period, China’s population increased rapidly, which made pressure on per capita food supply and the wider food system more serious. Official Chinese materials state that the country faced a long-term tension between a large population and limited farmland, and they also point to a second major birth peak from 1962 to 1972. Under these circumstances, population control became increasingly important by the end of the 1970s, and family planning policies were strengthened in response to wider development pressures. This thesis refers to this policy background only to explain why population growth became a key issue in the chosen period, rather than to provide a detailed policy analysis.

Nigeria and Egypt were chosen because both countries have experienced rapid population growth in recent decades and their food systems continue to face pressure. They are not treated as the same as China’s historical case. Instead, they are included because important recent trends—especially population growth and food supply pressure—make them suitable for comparison with China’s earlier experience. Therefore, in this thesis, China serves mainly as a reference case rather than a fully transferable model.

This thesis uses secondary national-level data rather than data from individuals, households, or regions. Therefore, the unit of analysis is the country-year. By comparing yearly

observations across the selected countries, the study identifies trends in population growth and food supply over time.

For China, the main period of analysis is 1961–1980, as it captures the relationship between population growth and food supply after the 1959–1961 crisis and before the long-term stabilization of food supply. Data from after 1980 are used in a supplementary way to illustrate broader improvement in food supply and the subsequent slowdown in population growth. For Nigeria and Egypt, the main period of analysis is 2010–2023, which is the most recent period with usable and comparable data on population and food supply, and it allows the study to examine whether recent trends may indicate possible future food-related challenges.

The three countries—China, Nigeria, and Egypt—were selected purposively rather than randomly because they match the aims of the research and comparable data are available for them. This approach allows a focused comparison between China’s earlier historical experience and the more recent situations in Nigeria and Egypt, while keeping the analysis centered on population growth and food supply.

This research sample is suitable for the aim of this thesis because it allows comparison across both different countries and different periods. China is used as a historical reference case, while Egypt and Nigeria are used as more recent cases. This helps the thesis examine whether similar population pressures may be connected with different food supply outcomes under different national conditions.

Another reason for choosing these three countries is that they all have large populations and important food-related challenges, but they do not have the same background. This makes the comparison more meaningful. At the same time, the thesis does not treat them as identical cases. Instead, it uses them to show how population growth and national food availability may develop differently across time and context.

4.2 Research method

This thesis uses a comparative descriptive research approach based on secondary quantitative data. It compares trends in population growth and food supply in China, Nigeria,

and Egypt across the selected periods. Instead of using advanced econometric models or causal tests, this research focuses on identifying and explaining general patterns in the data.

This research uses a comparative method by examining three different country cases—China, Nigeria, and Egypt—within one analytical framework. China serves as a historical reference case, while Nigeria and Egypt provide more recent cases. This comparison is used to show how population growth and food supply pressure developed under different conditions, and it also highlights broad similarities and differences among the three countries.

This research is descriptive in nature and relies on the direct presentation and interpretation of numerical data. Instead of carrying out tests on complicated statistical hypotheses or constructing forecasting models, this work investigates tendencies in the overall population, population growth rates, and food supply over time. These indicators are used to show how demographic pressure and food availability have changed within each country.

This thesis uses secondary data from two international databases: the World Bank and FAOSTAT. Population data are obtained from the World Bank, while food supply data come from FAOSTAT Food Balance Sheets. These sources were selected because they provide comparable national-level data and are widely used in academic and policy research.

The main variables used in the analysis are total population, population growth, and food supply (kcal/capita/day). Total population and population growth rates are treated as indicators of population pressure over time, while food supply (kcal/capita/day) is used as a simplified measure of national food availability. This thesis investigates the connection between these variables via direct contrast of their development tendencies, without employing advanced statistical estimation methods.

To support the analysis, the data are presented in tables and graphs. These visual materials make it easier to observe whether food supply improved, declined, or remained unstable during periods of rapid population growth. The main key focus of China is the time 1961–1980, hence for Nigeria and Egypt the primary time period is 2010–2023. The data of

China after 1980 are only used as supplementary reference materials to explain the long-term ascension of food supply and the subsequent deceleration of population growth.

This methodology is suitable for this current thesis because it does not make an attempt to explain every political, institutional, or economic cause of food insecurity. On the contrary, it gives a clear and concentrated comparison of how population growth and food supply have changed over time in the three chosen countries. Factors discussed in the existing literature—for example, water scarcity, imports, governance, conflict, and policy changes—are treated as background context rather than as equally developed empirical variables in this analysis.

The descriptive comparative method is appropriate for this thesis because the study is interested in general patterns rather than detailed causal testing. It helps show whether food supply improved, declined, or remained unstable during periods of population growth. This makes the results easier to present and compare across the selected cases.

This method also fits the limited scope of an undergraduate thesis. Instead of building a complicated model, it provides a simple way to compare changes in population and food supply over time. In this way, the research can remain focused while still giving meaningful results about food-related pressure in the three cases.

4.3 Research questions

Based on the research aim and the selected analytical framework, this thesis addresses the following research questions:

RQ1: How did population growth and food supply develop in China during 1961–1980, and how can this period be understood as a stage of food-related pressure under rapid demographic growth?

RQ2: How did food supply develop in China after 1980 in the context of slower population growth, and what does this reveal about the country's longer-term improvements in national food availability?

RQ3: How did population growth and food supply develop in Egypt and Nigeria during 2010–2023, and to what extent do these trends suggest growing food-related challenges in the future?

RQ4: What limited lessons can be drawn from comparing China’s historical case with recent population growth and food supply trends in Egypt and Nigeria?

4.4 Data collection

This thesis uses secondary quantitative data from two international databases: the World Bank and FAOSTAT. These sources provide publicly available and internationally comparable country-level data, which makes them suitable for a descriptive comparative study.

Population data are taken from the World Bank’s annual total population series, from which changes in population size and growth rates were calculated in order to support descriptive comparison. Food supply data were obtained from FAOSTAT Food Balance Sheets; the “Food supply (kcal/capita/day)” index was used as a national-level measurement for average food accessibility.

Data for the three selected countries—China, Nigeria, and Egypt—were collected according to the analytical periods used in this thesis. For China, the main period of analysis was 1961–1980, which matches the thesis’s historical focus, while data after 1980 were used only as supplementary references to show longer-term increases in food supply and the later slowdown in population growth. For Nigeria and Egypt, the main recent comparison period is 2010–2023.

After collection, the data were cleaned, standardized, and organized by country and year into separate datasets for population and food supply. From these annual data, annual changes, growth rates, and overall changes for the selected periods were calculated to make trend comparison easier. The organized data were then used to prepare tables, graphs, and descriptive comparisons for the analysis chapter.

Because this thesis examines national-level trends, it relies entirely on secondary data rather than primary data from individuals, households, or local regions. It does not construct an extensive multi-variable data collection that covers every aspect of food security. This research focuses on two core indicators—population and food supply—while using the literature review to place other related factors in the background, such as water scarcity, import reliance, governance, conflict, and policy changes.

This method of data collection is appropriate for the present research because it allows a direct and focused comparison of population pressure and national food supply across the selected countries and periods.

The use of the World Bank and FAOSTAT is important because both sources provide internationally comparable country-level data. This makes them suitable for a study that compares several countries across different periods. It also helps keep the empirical basis of the thesis clear and transparent.

The data collection process was designed to keep the comparison consistent. The same general types of data were collected for each country, namely population and food supply. This makes it easier to compare the three cases later in the analysis, because the thesis does not rely on different indicators for different countries.

4.5 Data analysis

This thesis uses a descriptive comparison of population and food supply trends in the selected countries and periods. Instead of attempting to prove a strict causal relationship, this analysis examines whether changes in population pressure were accompanied by increases, stagnation, or instability in food supply at the national level.

The analysis has been carried out by following several steps. First of all, the data were arranged in the order of time, according to country and year. For each country, trends in total population and food supply during the selected periods were examined and analyzed. Besides the original annual numerical values, we have calculated annual changes and growth rates, which can support clearer comparisons between different countries and in different time periods.

Second, the data have been put into tables and line diagrams to show the direction and speed of changes that happen to the population and food supply. In China, the main point of concentration is the 1961–1980 time period, which is regarded as an important historical stage for researching food-related pressure during a time of fast population growth. Data from after 1980 are only used as supplementary proof to display later enhancements in food supply and the deceleration of population growth. For Nigeria and Egypt, the analysis puts the focal point on the 2010–2023 time period, so as to inspect the recent development tendency of population pressure and the national food supply quantity.

Third, this research compares the first and final years of each selected time period in order to show the overall direction of change. This includes conducting examinations on increases in total population, changes in food supply in the same period, and whether improvements in food supply have the appearance to follow the speed of demographic growth. In this way, population growth is not treated as the only explanation for food-related problems, but as one important pressure factor that can be compared with changes in food availability.

To explain the level of food supply, this thesis takes 2,100 kcal/person/day as a simple reference standard for the lowest energy demand. This benchmark is used only as a practical reference point for descriptive discussion, and it does not automatically classify countries below or above it as food insecure or food secure. As discussed earlier, the FAO's method for measuring undernourishment does not use one universal calorie threshold, because minimum dietary energy requirements vary according to population structure. Hence, the 2,100 kcal reference standard is utilized herein solely as a straightforward reference point for wide-ranging alterations in national-level food supply capacity.

The analysis also includes a limited comparison among the three cases. In China, the data are used to show changes in food supply and population during a period of historical food-related pressure, followed by later improvement. As for Egypt and Nigeria, the recent data have been inspected by us to evaluate whether the continuous population growth has been accompanied by the promotion of food supply, or whether the trends show that the

pressure on food availability is continuing or increasing. This comparison is conducted with caution, because China is used as a historical reference case rather than a direct model for Egypt and Nigeria.

As discussed in the literature review and methodology, this thesis does not attempt to cover all possible causes of food insecurity or food production challenges. Other important elements, for example, water shortage, reliance on imports, management, conflict, and policy alteration, are recognized in the background part, but they are not studied with the same empirical meticulousness. Instead, the analysis focuses on two key indicators—population and food supply—in order to provide a clear and manageable comparison among the three countries.

Overall, this method of data analysis provides a simple way to examine changes in population pressure and national food supply over time in China, Egypt, and Nigeria, and to compare these trends across the selected periods.

The data analysis in this thesis focuses not only on the absolute level of food supply, but also on its direction of change over time. This is important because a country may remain above the benchmark and still show signs of increasing pressure if food supply weakens while population continues to grow. For this reason, the thesis pays attention to both the level and the trend of food supply.

Another important part of the analysis is the comparison between the selected cases. First, each country is examined separately in order to show its own population and food supply pattern. After that, the three cases are compared in a broader way. This makes it possible to identify both similarities and differences without treating the cases as the same.

5 Results

5.1 China's data

China, 1961–1980

In this thesis, the period 1961–1980 is treated as the core historical phase for examining the relationship between rapid population growth and food-related pressure in China. The results show that the population grew continuously during this period, while food supply also improved overall but remained comparatively low for most of the time and did not increase in a uniform way.

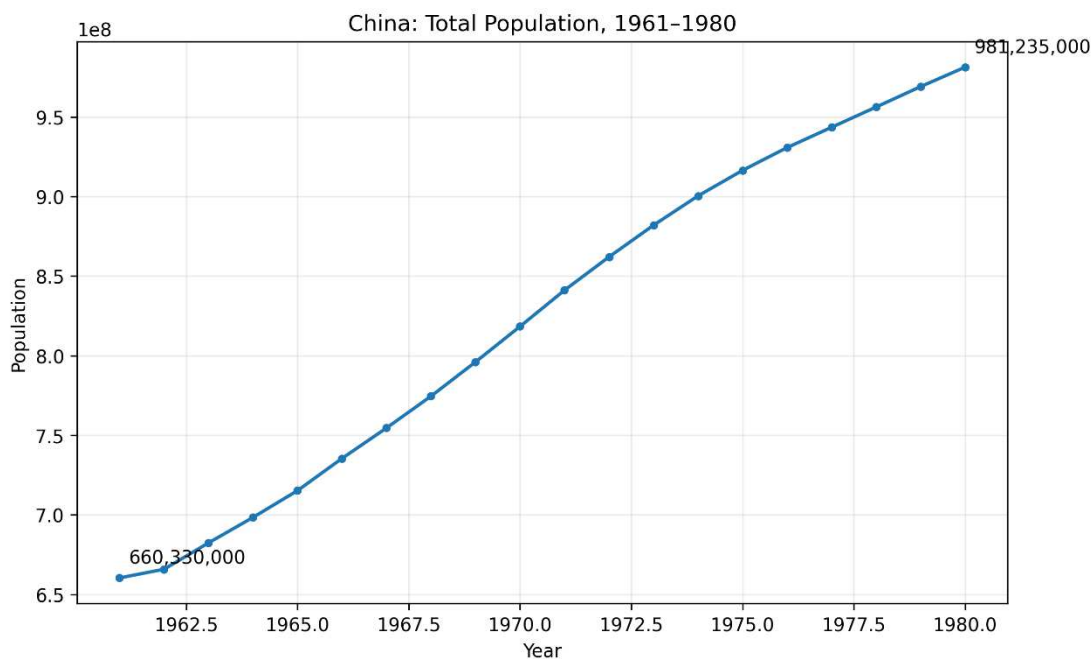


Figure 1 China: Total Population, 1961-1980

As Figure 1 shows, China's total population grew steadily from 660.33 million in 1961 to 981.24 million in 1980. The trend remained upward throughout the period, with no major decline. This means that China added more than 300 million people during these two decades, which suggests that population pressure remained strong throughout the whole period.

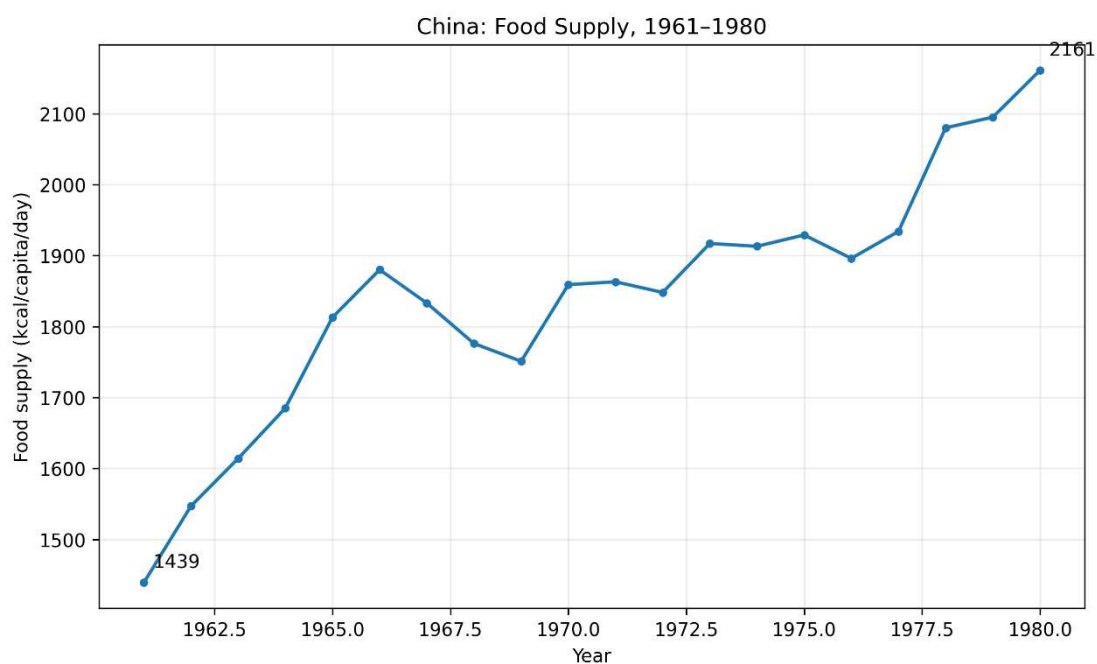


Figure 2 China: Food Supply, 1961-1980

At the same time, Figure 2 shows that China's food supply also improved during 1961–1980, rising from 1439 kcal/capita/day in 1961 to 2161 kcal/capita/day in 1980. However, unlike the population trend, the food supply trend was less stable. It rose quickly in the early 1960s, reached a higher level by the mid-1960s, then fluctuated through the late 1960s and most of the 1970s before a clearer increase appeared toward the end of the period. In other words, food supply improved overall, but the improvement was not steady.

Table 1 China: Key time points in population and food supply during 1961–1980

Year	Population (million people)	Food supply (kcal/capita/day)	Food supply vs 2,100 kcal
1961	660.33	1439	-661
1965	715.19	1813	-287
1970	818.32	1859	-241
1975	916.4	1929	-171
1980	981.24	2161	61

This kind of rule can be observed more clearly in Table 1, which gives some key time points. In the year 1961, the food supply was only 1439 kcal/capita/day, which is 661 kcal lower than the 2100 kcal reference point this thesis uses. Up to 1965, this value had increased to 1813 kcal/capita/day, but it was still 287 kcal lower than the reference standard. In 1970, it still stayed under the standard at 1859 kcal/capita/day, and in 1975, at 1929 kcal/capita/day, this number was still 171 kcal lower than the reference level. It was not until 1980 that food supply had a small rise, which got it above the benchmark, reaching 2161 kcal/capita/day, that is 61 kcal higher than 2,100.

The population increment in this same time interval also maintained a very strong status. According to Table 1 shows that the population quantity increased from 660.33 million in 1961 to 715.19 million in 1965, 818.32 million in 1970, 916.40 million in 1975, and 981.24 million in 1980. Hence, although food supply was still under the reference level in the majority of this time period, the population kept on having very fast growth.

Table 2 China: Interval changes in population and food supply during 1961–1980

Interval	Population absolute change (million)	Population average annual growth (%)	Food supply absolute change	Food supply average annual growth (%)
1961–1965	54.86	2.02	374	5.95
1965–1970	103.13	2.73	46	0.5
1970–1975	98.08	2.29	70	0.74
1975–1980	64.84	1.38	232	2.3

Table 2 gives the interval changes with more detailed manner. In the years 1961 to 1965, the total population had an increase of 54.86 million, while food supply rose by 374 kcal/capita/day. This was the time interval that had the most rapid food supply augmentation, possessing an average yearly growth rate of 5.95%. By way of comparison, in the

time period 1965–1970, the total population had an increase of 103.13 million, while food supply rose by only 46 kcal/capita/day, and it possessed an average annual growth rate that is 0.50%. A resembling mode emerged in 1970–1975, at which time the population got an increase of 98.08 million, but the food supply got an increase of merely 70 kcal/capita/day, with an average per year growth rate of 0.74%. Food supply at that time got clearer improvement in 1975–1980, increasing by 232 kcal per person per day, hence the population had an increase of 64.84 million.

Put all together, these outcomes indicate that China in 1961–1980 underwent continuous and significant population growth, together with a comparatively limited food supply for the majority of this period. Although the overall food supply got better, it still stayed under the 2,100 kcal/person/day reference level for the majority of the studied years, and its increase was not even among the different time periods. In the same period, the population continuously grew, which shows that pressure on national food supply remained strong.

On the whole, the evidence that comes from Figure 1, Figure 2, Table 1 and Table 2 supports that 1961–1980 can be treated as a historical stage of food connection pressure in the time of rapid population growth. Food supply has indeed obtained improvement along with time, but this improvement was gradual and uneven, and it only moved a little higher than the benchmark up to 1980.

Another point that can be observed from these results is that the improvement in food supply during 1961–1980 should be interpreted with caution. Although the overall trend was upward, the starting level was very low, and food supply remained below the 2,100 kcal/capita/day reference point for most of the period. This means that the rise in food supply did not immediately remove the pressure created by rapid population growth. Instead, the results suggest that improvement took place gradually and under conditions of continuing demographic expansion.

In addition, the interval data show that progress was not evenly distributed across the whole period. The most obvious increase in food supply appeared in 1961–1965, while the later intervals of 1965–1970 and 1970–1975 showed much smaller gains. This uneven

development is important because it indicates that national food availability did not strengthen in a stable way while the population continued to grow quickly. For the purposes of this thesis, this supports the interpretation of 1961–1980 as a period in which food-related pressure remained present even though food supply improved overall.

China, 1981–2023

In this thesis, we take the time after 1980 as additional proof to display how China's population and food supply developed after the early period of pressure connected to food. When we compare with 1961–1980, this later time period displays a dissimilar mode: population did keep on growing, but it grew at a slower speed, hence food supply had an obvious increase, and it was always above the 2,100 kcal/capita/day reference point during all the time.

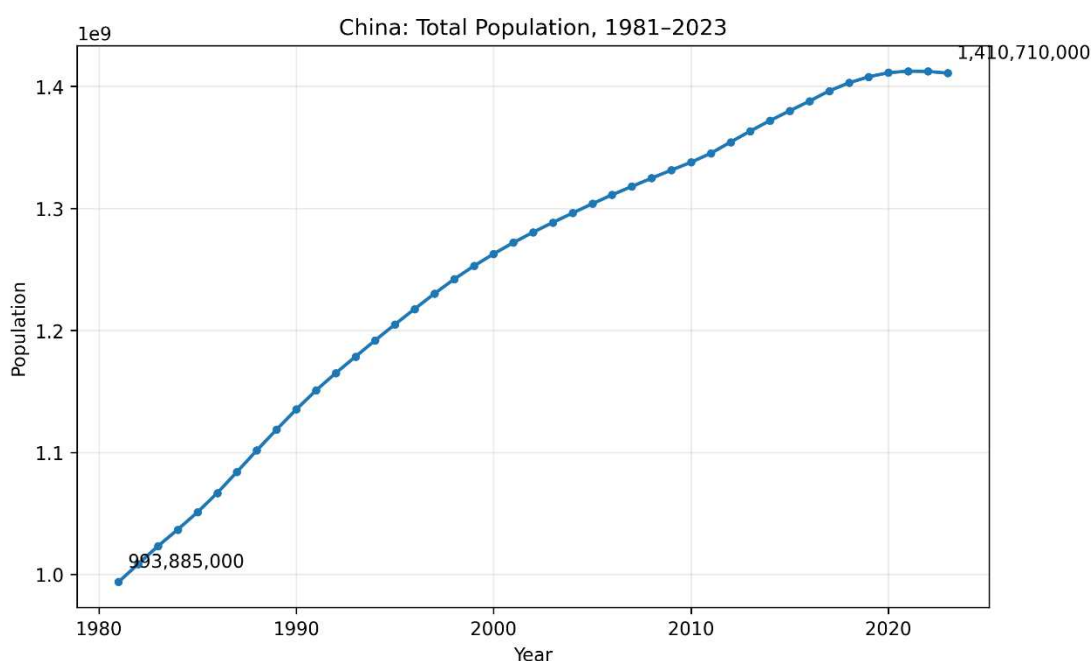


Figure 3 China: Total Population, 1981–2023

Just like what Figure 3 has displayed, the total population of China, from 993.89 million in 1981, went up to 1,410.71 million in 2023. Although the population still keeps a very large size in absolute terms, the trend of its growth is smoother and not so steep, hence it differs from the earlier period. In the later time period, the curve gets flatter, which indicates that population growth gradually became slower.

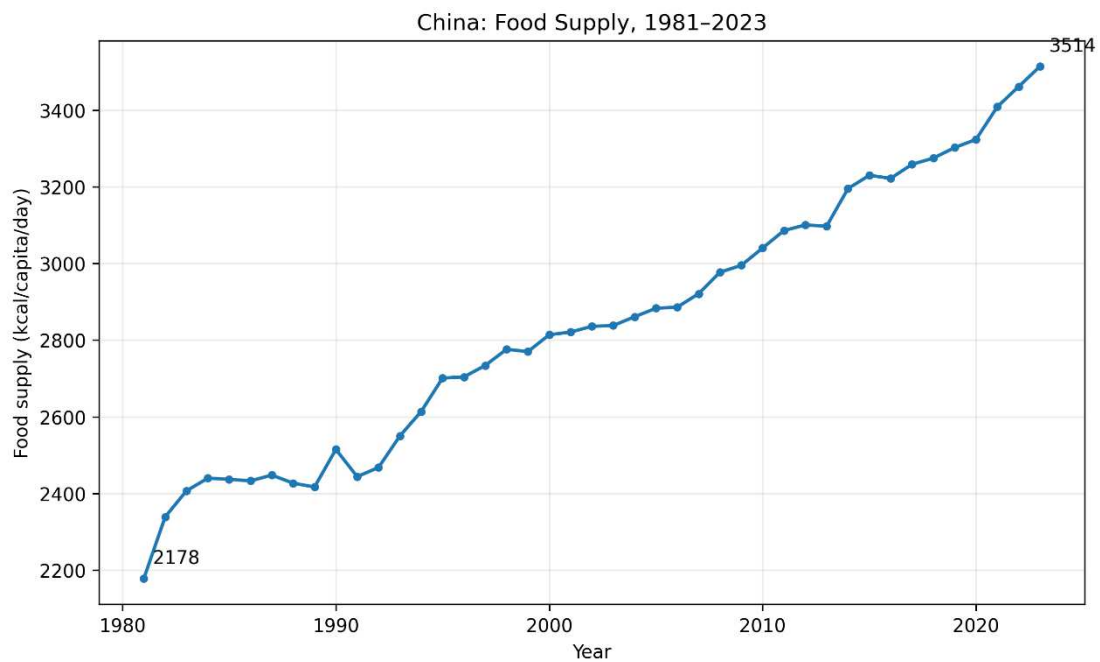


Figure 4 China: Food Supply, 1981–2023

In the meantime, Figure 4 has manifested an obvious long-term ascending trend in food supply. The food supply quantity increased from 2178 kcal/capita/day in 1981 to 3514 kcal/capita/day in 2023. Different from the time before, the trend after 1980 remained all the time above the 2,100 kcal/capita/day reference point, which this thesis uses. Although we still can see some short-term fluctuations, the overall tendency is going upward, and it is more stable than the period of 1961–1980.

Table 3 China: Key time points in population and food supply during 1981–2023

Year	Population (million people)	Food supply (kcal/capita/day)	Food supply vs 2,100 kcal
1981	993.89	2178	78
1990	1135.19	2515	415
2000	1262.65	2814	714
2010	1337.71	3040	940
2020	1411.1	3323	1223
2023	1410.71	3514	1414

This kind of arrangement is also displayed in Table 3, which gives data on a number of important time points. In the year 1981, the food supply had already been 78 kcal higher than the 2,100 kcal benchmark. It therefore then further increased to 2515 kcal/capita/day in 1990, 2814 kcal/capita/day in 2000, 3040 kcal/capita/day in 2010, 3323 kcal/capita/day in 2020, and 3514 kcal/capita/day in 2023. When we make a comparison with the standard reference, the difference becomes larger, from +78 in the year 1981 to +1414 in the year 2023. This shows that an obvious enhancement of the nation-level food supply ability exists in the time after 1980.

The population was still big and kept on increasing in this time period, but the speed got more slow as time went by. According to Table 3 shows, the total population number was 993.89 million in the year 1981, 1135.19 million in the year 1990, 1262.65 million in the year 2000, 1337.71 million in the year 2010, 1411.10 million in the year 2020, and 1410.71 million in the year 2023. These data display long-term spreading, but they also point out that population growth got more and more gentle in the following decades.

Table 4 China: Interval changes in population and food supply during 1981–2023

Interval	Population absolute change (million)	Population average annual growth (%)	Food supply absolute change	Food supply average annual growth (%)
1981–1990	141.3	1.48	337	1.61
1990–2000	127.46	1.06	299	1.13
2000–2010	75.06	0.58	226	0.77
2010–2020	73.4	0.54	283	0.89
2020–2023	-0.39	-0.01	191	1.88

This speed decrease can be seen more clearly in Table 4. In the period of 1981 to 1990, the population increased by 141.30 million, with an average annual growth rate being 1.48%. In the period of 1990 to 2000, this quantity increased by 127.46 million, hence it

has an average annual growth rate of 1.06%. In the period of 2000 to 2010, the increment attained 75.06 million, possessing an average per year growth rate of 0.58%, and in the period of 2010 to 2020, this number was 73.40 million, possessing an average per year growth rate of 0.54%. At last, in the period from 2020 to 2023, what this table shows is a tiny decrease of 0.39 million, which has an average every year growth rate of -0.01%. This illustrates that population growth has turned into a rate that is much slower than the one in the past, in the earlier historical stage.

Food provision, however, continuously got better in these same time periods. According to Table 4, it has an increment of 337 kcal/capita/day in the time period 1981–1990, 299 kcal/capita/day in the time period 1990–2000, 226 kcal/capita/day in the time period 2000–2010, 283 kcal/capita/day in the time period 2010–2020, and 191 kcal/capita/day in the time period 2020–2023. The corresponding average yearly food supply increase speeds were 1.61%, 1.13%, 0.77%, 0.89%, and 1.88%. Therefore, food supply was continuously increasing throughout the whole time after 1980, even though the speed of growth was not the same in different time periods.

Putting all together, the outcomes from 1981 to 2023 exhibit an obvious change from the earlier mode that was depicted in the first portion of the China part. In the previous time stage, food provision remained lower than the standard value for the majority of the time; meanwhile, the population growth was very fast. In the time after 1980, the food supply was kept above the standard line all the time, thus the population increase slowly got slower. Therefore, the results after 1980 display that a stage of raised national food supply exists, together with population growth that is slower.

On the whole, the evidence that comes from Figure 3, Figure 4, Table 3 and Table 4 indicates that the development of China after 1980 has a clear difference compared with its earlier historical experience. The population was still large, but its speed of growth became slower as time went by, while the provision of food increased by a big margin and remained distinctly higher than the 2,100 kcal/capita/day reference point which this thesis uses.

The results for the period after 1980 also show that improvement in food supply was not simply a short-term change, but a long-term pattern. Food supply remained above the benchmark throughout the whole period and the gap above the benchmark widened considerably by 2023. This suggests that the post-1980 stage was not only different from the earlier period in relative terms, but also stronger in absolute terms, because the country maintained a much higher level of national food availability over several decades.

At the same time, the population trend in this later period should not be understood as the absence of demographic pressure. China still had a very large and growing population for most of the post-1980 period. However, compared with the earlier stage, the pace of growth became more moderate, and this occurred together with clearer improvement in food supply. In this thesis, this combination is important because it shows a different empirical relationship between population and food availability from the one observed during 1961–1980.

Final summary of the Chinese case

Table 5 China: Overall comparison of population and food supply across two periods

Period	Indicator	Start year	End year	Start value	End value	Absolute change	Average annual growth (%)
Pressure stage (1961–1980)	Population (million people)	1961	1980	660.33	981.24	320.91	2.11
	Food supply (kcal/capita/day)	1961	1980	1439	2161	722	2.16
	Food supply vs 2,100 kcal	1961	1980	-661	61	—	—
	Population (million people)	1981	2023	993.89	1410.71	416.83	0.84

Improvement stage (1981–2023)	Food supply (kcal/capita/day)	1981	2023	2178	3514	1336	1.15
	Food supply vs 2,100 kcal	1981	2023	78	1414	—	—

Table 5 makes a comparison of the main outcomes from the two time periods and displays an obvious discrepancy between China's earlier and later development stages. In the 1961–1980 pressure phase, population quantity grew from 660.33 million to 981.24 million, an absolute increment of 320.91 million, which has an average annual growth rate of 2.11%. In this identical time frame, the food provision went up from 1439 kcal/capita/day to 2161 kcal/capita/day, which is an absolute increment of 722 kcal/capita/day, with an average yearly growth rate of 2.16 percent. However, on account of food supply commenced from a very low starting level, it still stayed under the 2,100 kcal reference standard for the majority of the time period, moving from -661 relative to the reference standard in 1961 to merely +61 in 1980.

By comparison, in the 1981–2023 promotion stage, China still possessed an extremely huge population, and population growth has become much slower. The population quantity rose from 993.89 million in 1981 to 1410.71 million in 2023, a total absolute growth of 416.83 million; the average every year growth rate dropped to 0.84%. In this identical time interval, the food provision quantity rose from 2178 kcal/capita/day to 3514 kcal/capita/day, which is an absolute growth amount of 1336 kcal/capita/day, and it has an average yearly increase rate of 1.15 percent. Being different from the time before, food supply has remained over the 2,100 kcal reference level throughout this time; the gap is growing from +78 in 1981 to +1414 in 2023.

Putting all together, these outcomes display discrepancies between the two time stages not merely in the degree of food supply, but also in the more wide-ranging connection between population pressure and the amount of usable food. In the time that came before, the quick growth of population was put together with comparatively small food provision, and the amount of food people could get stayed near or lower than the standard for the

greater part of the years that were studied. In the subsequent stage, the growth of population became slower, whereas the supply of food got further better and kept a much bigger gap above the reference standard.

Hence, the Chinese example in this thesis can be split into two distinct experiment stages. The first time period, 1961–1980, had as its mark powerful population pressure and a relatively restricted food supply. The second stage, which is 1981–2023, has exhibited more distinct enhancement in the nationwide food supply condition under the condition of slower population increase. This difference possesses significance for the current research, as it offers the historical reference point for the subsequent comparison with Egypt and Nigeria.

Seen together, the two stages of the Chinese case highlight that the key difference lies not only in whether food supply rose, but in the broader balance between food availability and demographic pressure. In the earlier stage, food supply improved from a low base and remained near or below the benchmark for most of the period, while population growth stayed rapid. In the later stage, food supply improved from a much stronger starting point and remained clearly above the benchmark, while population growth slowed. This contrast is central to the present thesis because it shows how the same country can display two clearly different patterns over time.

For that reason, China is especially useful in this study as a historical reference case. It provides an empirical example of a shift from a period of stronger food-related pressure to a later period of clearer improvement. This does not mean that other countries will necessarily follow the same path. However, it does make the Chinese case valuable for comparison, because it helps show why continued population growth should be considered together with the direction and level of food supply.

5.2 Egypt's data

In this thesis, we have taken Egypt as a new case to carry out an examination of the relation between continuous population growth and national food supply from 2010 to 2023. It is different from China's situation in history; Egypt's food supply did not reach

the degree of critical shortage in terms of absolute total calories. However, the outcomes display a dissimilar category of pressure: when the population was growing in a stable way, food provision did not get better in the same period and instead went down on the whole, hence it had some undulations.

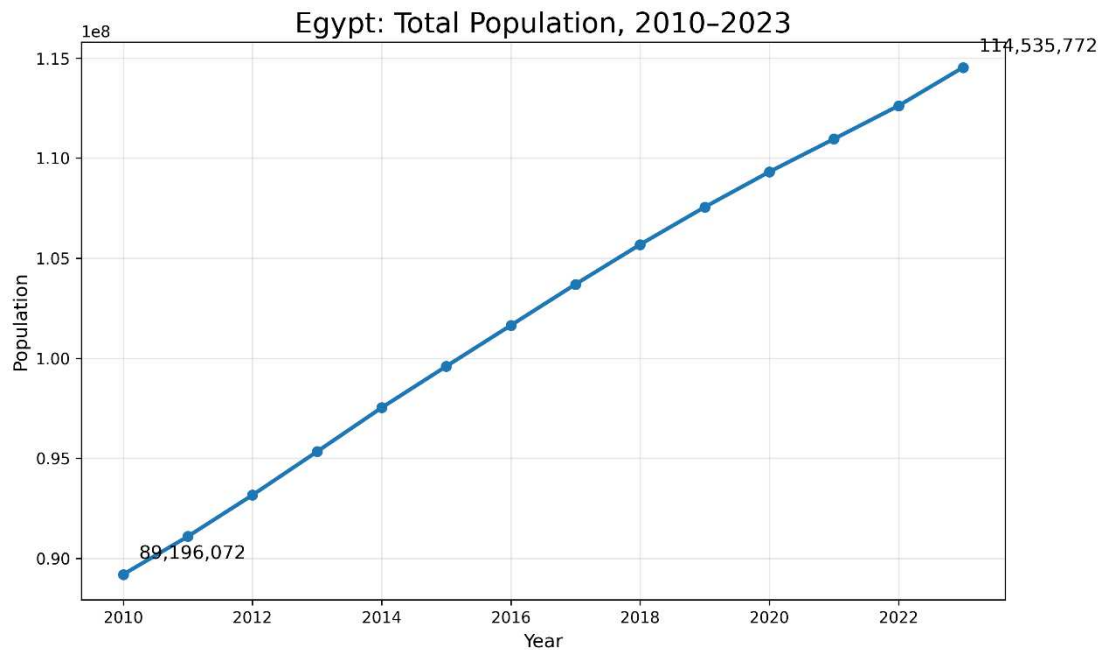


Figure 5 Egypt: Total Population, 2010–2023

According to what Figure 5 displays, the total population of Egypt had a steady increase, from 89.20 million in the year of 2010 to 114.54 million in the year of 2023. This tendency has been going up, and there was no drop in this time period. In these thirteen years, this country has increased by over 25 million persons, which shows that there is continuous population pressure upon the nation's resources and the need for food.

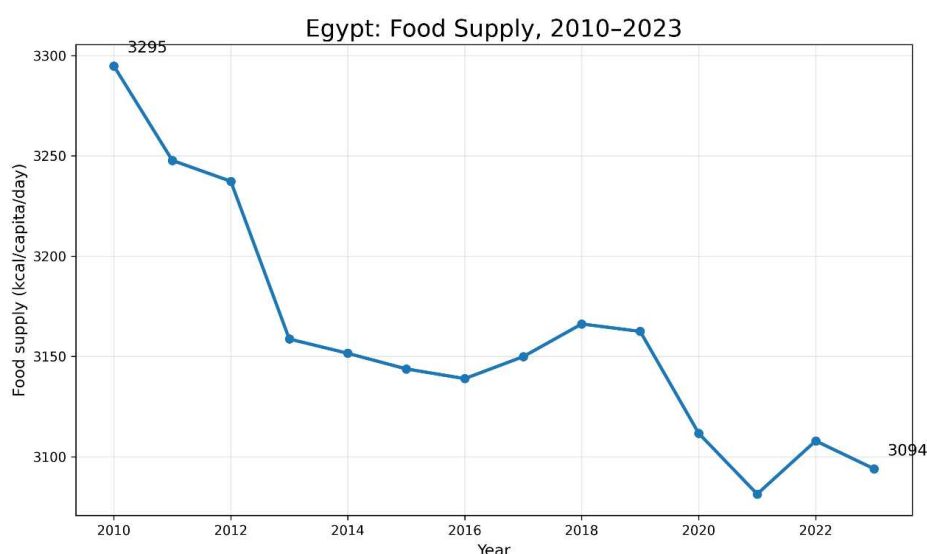


Figure 6 Egypt: Food Supply, 2010–2023

On the opposite side, Figure 6 lets us see that the food supply of Egypt has followed a trend that is different. In the year 2010, this value began at 3295 kcal/capita/day, and in the whole period it tended to decrease, although there existed some fluctuations which were short-term, and by 2023 it had reached 3094 kcal/capita/day. When we compare with China's period after 1980, Egypt has not experienced a distinct long-run promotion in the aspect of food provision. On the contrary, the absolute quantity of food supply is still kept at a relatively high level, but as a whole, it has a downward trend.

Table 6 Egypt: Key time points and interval changes in population and food supply

Year	Population (million people)	Population absolute change (million)	Population average annual growth (%)	Food supply (kcal/capita/day)	Food supply vs 2,100 kcal	Food sup- ply aver- age an- nual growth (%)	Interval
2010	89.2	—	—	3295	1195	—	2010 baseline
2015	99.6	10.4	2.23	3144	1044	-0.93	2010– 2015

2020	109.32	9.72	1.88	3112	1012	-0.2	2015– 2020
2023	114.54	5.22	1.57	3094	994	-0.19	2020– 2023

This kind of pattern may also be observed in Table 6, which summarizes key time points and changes in intervals. The food supply of Egypt in 2010 was 3295 kcal/capita/day, which was 1195 kcal higher than the 2100 kcal reference point that this thesis uses. It afterwards dropped to 3144 kcal/capita/day in 2015, which is 1044 kcal higher than the benchmark. It declined still further to 3112 kcal/capita/day in 2020, that is 1012 kcal higher than the benchmark; it arrived at 3094 kcal/capita/day in 2023, which was 994 kcal higher than the benchmark.

These data indicate that Egypt's food provision stayed over the 2,100 kcal reference value during the entire period. However, the difference that is above the standard reference became smaller as time went by. In other words, the country-level food supply capacity did not suffer a total breakdown, yet it experienced a weakening rather than an enhancement in the period from 2010 to 2023.

Population, on the opposite side, kept on going up steadily through all important years. This form can show that Egypt's population, from 89.20 million in the year 2010, has risen to 99.60 million in 2015, 109.32 million in 2020, and 114.54 million in 2023. Hence, the growth of population kept being continuous in the whole of this time, even though the food supply presented an overall decreasing tendency.

The data of the interval make this correlational relation even more manifest. In the years from 2010 to 2015, the total population increased by 10.40 million, with an average annual growth rate of 2.23%, at the same time, the food supply had a decline of 151 kcal/capita/day, with an average per year change of -0.93%. During 2015 to 2020, the population increased by another 9.72 million, holding an average each year growth rate of 1.88%; meanwhile, food supply decreased by another 32 kcal/capita/day, holding an average annual change of -0.20%. In the time from 2020 to 2023, the total population increased by

5.22 million, holding a per year average growth rate of 1.57%; meanwhile, food supply decreased by 18 kcal/capita/day, holding a per year average change of -0.19%.

Putting all things together, these results show an obvious comparison between population development tendencies and food supply in Egypt during the period 2010–2023. In the entire time period, the population showed a steady increase, but the total quantity of food supply had a general decline. Although the food supply still stayed at a level much higher than the 2,100 kcal standard benchmark, it did not get better together with the growth of the population. On the contrary, the difference above the reference standard has narrowed, from +1195 kcal in 2010 to +994 kcal in 2023.

On the whole, the proof coming from Figure 5 and Figure 6, together with the matching table about Egypt, hence gives us the indication that this nation's recent developing mode ought not to be comprehended as a type of seriously low caloric food supply. On the contrary, it demonstrates that the population grows continuously, and at the same time, the food that the whole nation can get is becoming weaker. This kind of pattern is also in accordance with the documents we reviewed before, which hold that the food security challenges of Egypt are connected not only to the total amount of food, but also to structural elements such as restriction of resources, dependence on importing, and being easily harmed by outside impacts.

One important point in the Egyptian case is that the problem shown by the data is not one of extremely low food supply in absolute terms. Throughout the whole period, food supply remained clearly above the 2,100 kcal/capita/day benchmark used in this thesis. However, the direction of change is less favorable. Instead of improving alongside population growth, food supply weakened over time. This means that the Egyptian case is important not because it reflects immediate caloric shortage, but because it shows that a relatively high national food supply can still move in a negative direction under continued demographic growth.

The narrowing gap above the benchmark is also significant in descriptive terms. In 2010, Egypt's food supply stood far above the reference point, but by 2023 the surplus had become smaller. Although this reduction does not in itself indicate a food crisis, it does

suggest that national food availability became less favorable over time. In the context of this thesis, such a pattern deserves attention because it may indicate that demographic growth is not being matched by equally strong improvement in food supply.

For the goal of this thesis, Egypt's example hence shows that lasting pressure connected with food exists when population grows quickly, but this form is not the same as China's experience. While China in 1961–1980 put together fast population growth with a relatively small food supply for most of this period, Egypt in 2010–2023 had continuous population growth together with a falling food supply trend that stayed above the standard level but got no better over time.

The interval data also strengthen this interpretation because they show that the overall decline in food supply was not limited to only one short period. Food supply fell during 2010–2015, continued to decline during 2015–2020, and decreased again during 2020–2023. At the same time, population kept rising in all three intervals. This consistency matters because it suggests that the Egyptian pattern was not simply the result of one isolated fluctuation, but rather a broader tendency across the whole study period.

For the present thesis, Egypt therefore represents a case in which food-related pressure appears in a more gradual form. The country did not experience the very low food supply levels seen in China's earlier historical stage, yet the combination of rising population and weakening food supply still raises concern. In this sense, Egypt's results contribute to the broader argument of the study that food-related pressure should be considered not only in terms of absolute levels, but also in terms of whether national food availability is improving or deteriorating over time.

5.3 Nigeria's data

In this thesis, we take Nigeria as a recent example to study the connection between quick population growth and national food supply during the period from 2010 to 2023. When we make the comparison between Egypt and Nigeria, Nigeria has the experience of more rapid population growth in this period of time. If we make a comparison with the experience of China after 1980, this place has not displayed an obvious long-term enhancement

in the domain of food supply. On the contrary, the outcomes show that when the population went up without stopping and greatly, food provision changed up and down and on the whole got lower by the end of this time.

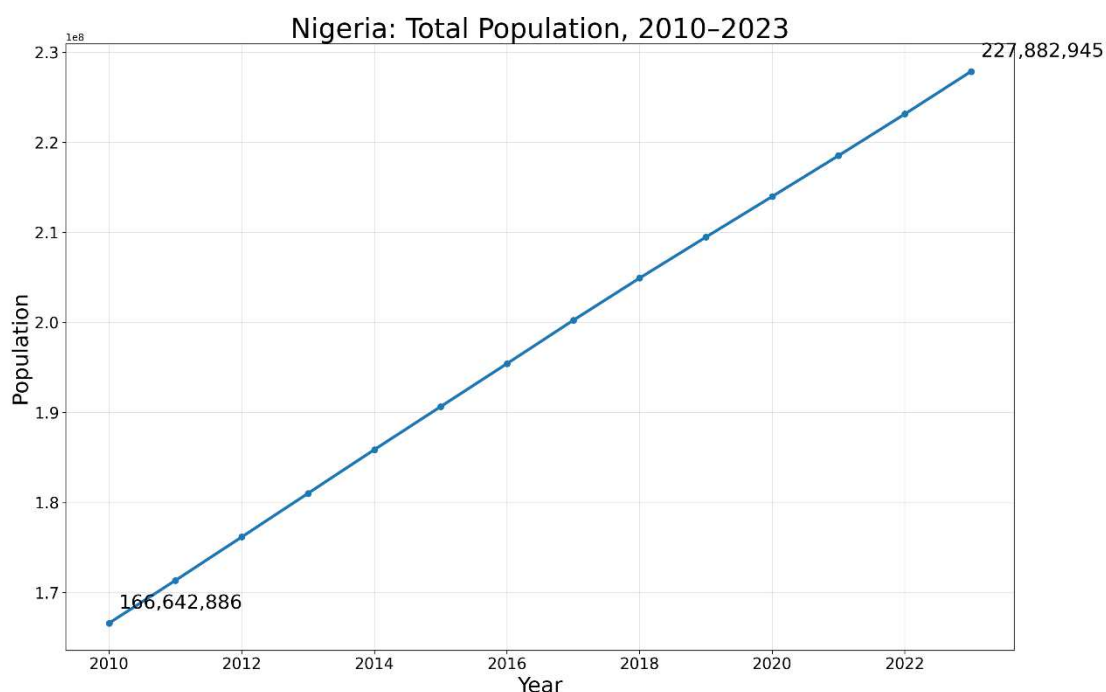


Figure 7 Nigeria: Total Population, 2010–2023

According to what Figure 7 shows, the entire population of Nigeria has a steady rise; it went from 166.64 million in the year 2010 to 227.88 million in the year 2023. This tendency maintained a continuous rise; it did not have any fall during this time span. In these thirteen years, this country has increased by more than 61 million people, which shows very big population pressure and a quick rising of food demand.



Figure 8 Nigeria: Food Supply, 2010–2023

By comparison, Figure 8 displays that the food supply of Nigeria did not follow the same kind of positive developing trend. Its value was 2566 kcal/capita/day in the year 2010, and although there are changes from year to year and several short-period increases, it fell to 2470 kcal/capita/day in 2023. Being different from the population trend that rises continuously, the food supply has no stability, and hence finishes this time period at a level that is lower than the level where it started.

Table 7 Nigeria: Key time points and interval changes in population and food supply

Year	Population (million people)	Population absolute change (million)	Population average annual growth (%)	Food supply (kcal/capita/day)	Food sup- ply vs 2,100 kcal	Food sup- ply aver- age annual growth (%)	Interval
2010	166.64	—	—	2566	466	—	2010 baseline
2015	190.67	24.03	2.73	2560	460	-0.05	2010– 2015

2020	214	23.32	2.33	2497	397	-0.5	2015– 2020
2023	227.88	13.89	2.12	2470	370	-0.36	2020– 2023

This kind of rule can also be seen in Table 7, which sums up key time points and changes between intervals. The food supply of Nigeria in 2010 was 2566 kcal per person per day, which was 466 kcal higher than the 2,100 kcal benchmark that is utilized in this thesis. It afterward decreased a little to 2560 kcal/capita/day in 2015, that is 460 kcal higher than the benchmark, hence it dropped still more to 2497 kcal/capita/day in 2020, that is 397 kcal higher than the benchmark, thus it arrived at 2470 kcal/capita/day in 2023, which was 370 kcal higher than the benchmark.

These data indicate that Nigeria's food supply stayed over the 2,100 kcal reference level for the entire duration of the period. However, the gap that is above the benchmark became smaller over time. In other words, the food that the whole nation can get did not drop below the standard level, but it became weaker instead of getting better in the time period from 2010 to 2023.

Population, on the other hand, was continuously rising with great force in all the important years. What the table displays is that Nigeria's population had a growth from 166.64 million in 2010 to 190.67 million in 2015, 214.00 million in 2020, and 227.88 million in 2023. Therefore, population growth still kept continuous and big in the whole of this period, even when food supply on the whole had a drop.

The interval data can make this relationship even clearer. In the period from 2010 to 2015, the total population increased by 24.03 million, having an average yearly growth rate of 2.73%, and at the same time, the food supply decreased by 6 kcal/capita/day, hence having an average yearly change of -0.05%. In the period from 2015 to 2020, the total population had a further increase of 23.32 million, possessing an average annual growth rate of 2.33%, and at the same time, food supply had an additional decrease of 63 kcal/capita/day, having an average annual change of -0.50%. In the period from 2020 to 2023, the

total population got an increase of 13.89 million, with an average each year growth rate of 2.12%; meanwhile, food supply decreased by 27 kcal/capita/day, with an average each year change of -0.36%.

After putting all these things together, these results give a quite clear comparison of the change trends of population and food supply in Nigeria in the 2010–2023 time period. The population in this whole time period experienced a fast and uninterrupted growth; meanwhile, the food supply had changes between high and low, but on the whole it showed a fall. Although food supply still stayed higher than the 2,100 kcal benchmark, it did not get better in accordance with population growth. On the contrary, the extra amount that is higher than the reference standard has decreased from +466 kcal in the year 2010 to +370 kcal in the year 2023.

All in all, the proof coming from Figure 7 and Figure 8, together with the attached form that concerns Nigeria, indicates that this nation's recent developing mode should not be comprehended as a type of extremely low-calorie food supply. On the contrary, it demonstrates that under the condition of vigorous population growth, the continuous pressure related to food has not been matched by the promotion of the national food supply level. This kind of mode is also in accordance with the documents we looked over before, which point out Nigeria's rapid population growth, bad agriculture achievement, and structural weaknesses.

The Nigerian case is especially notable because the demographic trend is stronger than in Egypt during the same period. While both countries experienced continuous population growth, Nigeria added a much larger number of people and maintained higher average annual growth rates across the intervals. At the same time, food supply did not show long-term improvement. Instead, it fluctuated and ended the period at a lower level than in 2010. This combination makes the Nigerian pattern particularly important in the present thesis, because it suggests a stronger form of demographic pressure without corresponding strengthening in national food availability.

Another important feature is that the surplus above the 2,100 kcal/capita/day benchmark was much smaller in Nigeria than in Egypt even at the beginning of the period, and this

surplus narrowed further by 2023. This means that Nigeria not only experienced continued population growth together with declining food supply, but also did so with a smaller margin above the benchmark. In descriptive terms, this makes the Nigerian case more concerning within the limited analytical framework of this thesis.

For the purposes of this thesis, the case of Nigeria shows continued food-related pressure under conditions of rapid population growth. In contrast to China's later period, when slower population growth was accompanied by strong improvement in food supply, Nigeria shows a different pattern: rapid population growth together with unstable and, overall, declining food supply. Compared with Egypt, Nigeria shows a similar problem in that food supply did not improve over time, but its population pressure was even greater, as reflected in the consistently higher population growth rates during the period.

The interval pattern also supports the view that the Nigerian case was marked by continuing pressure rather than temporary change. Food supply did not collapse sharply in one single interval, but it weakened across the study period while the population continued to rise strongly. This is important because it suggests that the problem was not simply short-term instability. Rather, the results point to a broader pattern in which food availability did not keep pace with continuing demographic expansion.

For the purpose of comparison in this thesis, Nigeria can therefore be understood as the most demographically pressured of the two recent African cases. Unlike China after 1980, it did not combine slower population growth with stronger food supply performance. Unlike Egypt, it also showed a smaller surplus above the benchmark and stronger population growth rates. For this reason, the Nigerian results provide important support for the argument that rapid and sustained demographic growth may become more difficult to manage when national food supply does not improve at the same time.

5.4 Discussion

Taken together, the results from China, Egypt, and Nigeria show that the core concern of this thesis is not only whether a country's food supply is above or below the 2,100

kcal/capita/day benchmark, but also how food supply changes under continued population growth. In this thesis, the three cases are examined within one shared analytical framework: population growth is used as an indicator of demographic pressure, while food supply is used as a simplified measure of national food availability. From this perspective, the three countries show different patterns. During 1961–1980, China experienced rapid population growth together with relatively limited and uneven food supply. After 1980, China showed slower population growth together with clearer long-term improvement in food supply. By contrast, between 2010 and 2023, both Egypt and Nigeria experienced sustained population growth without corresponding improvement in food supply. This does not mean that the three countries faced the same type of food security problem, because food security is broader than food availability alone and also includes access, stability, and utilization. However, within the limited scope of this thesis, the comparison suggests that continued population growth becomes more concerning when national food supply does not improve at the same time, or when the gap above the benchmark becomes smaller.

China holds a particularly important place in this discussion because, within this thesis, it is used as a historical reference case rather than as a direct model for Egypt and Nigeria. The value of the Chinese case lies in the fact that it shows two clearly different stages within one country. The first period, 1961–1980, was marked by rapid population growth together with relatively low and uneven food supply, which created continuous pressure on national food availability. This pattern generally agrees with the literature reviewed earlier, which links China's food problems in the late 1950s, the 1960s, and the early 1970s not only to agricultural weakness and institutional problems, but also to strong population pressure and limited per capita food availability. The second stage, which is after 1980, has manifested a dissimilar mode, with slower population growth and clearer long-term advancement in food supply. For the aims of this thesis, this two-step mode renders China analytically valuable, as it assists in making a distinction between a stage of pressure and a later stage of promotion within the same national case. At the same time, this point therefore does not mean that China can give a simple answer for the African cases. On the contrary, it provides a historical reference which helps to explain why, when

we make comparisons, the continuing growth of population together with the bad performance of food supply is something that ought to be paid attention to.

However, Egypt and Nigeria should not simply be regarded as “the China of today.” Although the comparison used in this thesis is useful, the three cases differ in important ways, which makes direct equivalence unsuitable. As discussed in the literature review, China’s early food-related pressure was connected to a particular historical background that included famine, central planning mistakes, agricultural weakness, and later major institutional and production changes. By comparison, the literature on Egypt emphasizes structural constraints such as water scarcity, dependence on irrigation, reliance on imports, and vulnerability to external market shocks. The literature on Nigeria, by contrast, emphasizes long-term food poverty, rapid urban development, weak agricultural performance, governance problems, conflict, and climate-related pressure. These differences matter because they show that similar pressure on national food supply does not necessarily come from the same causes. For this reason, the present research does not claim that Egypt and Nigeria are directly repeating China’s experience. Instead, the comparison has a more limited purpose: it shows that when population continues to grow but food supply does not improve correspondingly, pressure on national food availability can remain even when the wider historical and structural conditions are very different.

In this restricted comparative frame, the outcomes of Egypt and Nigeria may yet be regarded as caution signals for possible coming food production challenges. This is not because, at present, either of these two nations has the same type of low food provision level that was observed in China in the 1961–1980 time period. Actually, the two nations have always kept above the 2,100 kcal standard throughout the whole research time. A more significant issue that we must discuss is the orientation of change, when we consider food supply together with the increment of population. In the country of Egypt, the population total went up continuously while the provision of food decreased on the whole, and the gap beyond the standard reference became narrower. From a demographic perspective, in Nigeria, the condition was even more concerning because the growth of the population was more rapid, and the ultimate food supply was also lower than that which existed at the beginning of the research period. In this thesis, we give a careful explanation

of these trends as warning signals, because they point out that the national food supply condition is not getting better, along with continuous population growth. This cannot prove that a future food crisis will happen, and it also cannot include all the factors that determine the safety of food. However, this thing does give support to the argument, which is that if population pressure keeps being big while food supply does not grow or becomes smaller, food production challenges may be more difficult to control as time goes by.

This explanation, however, must be comprehended within the restrictions of this current research. First, this thesis depends upon macro-level secondary data, and therefore, it only inspects wide national trends instead of regional differences, household situations, or individual food acquisition. As we pointed out before, the food supply on country level is not equal to the real food people eat, and it thus cannot reflect the unequal problems of food allocation inside one nation. Second point, the said analysis is of a descriptive nature but not causal. This result indicates that population growth and food supply have changed following certain rules through time, but it is not able to prove that one of them directly produces the causation to the other. Third, many important key factors that have been pointed out in existing documents were not subjected to the same degree of empirical investigation in the current research; this includes governance, conflict, trade dependence, water shortage, technology change, agriculture policy, and more extensive economic shocks. These leave-outs are especially of great importance in the situations of Egypt and Nigeria, hence the existing research documents make it clear that pressure connected with food is formed by many structural conditions, therefore, it is not formed by demographic growth alone. At the end, the utilization of the 2,100 kcal/person/day standard is merely a simplified reference point that is used for descriptive comparison. This method has usefulness for displaying wide changes in the amount of food that can be obtained, but therefore it must not be regarded as a full measurement of food safety or as a rigorous standard line for evaluating the situations of whole countries.

Under these restrictions, the Chinese example still provides certain restricted experiences for comparison. The core enlightenment is not that Egypt or Nigeria ought to walk a par-

ticular Chinese road, nor that China offers a universal pattern. On the contrary, the meaning of the China example inside this dissertation is that it historically demonstrates that strong population growth, when it is combined with weak or unbalanced progress in food obtainable, can be connected with continuous pressure upon the food system, and therefore how a later time period with slower population growth and stronger food supply achievement looks different from the perspective of empirical data. This comparison makes clear that recent developments in Egypt and Nigeria deserve attention. It also suggests that, in comparative analysis, the core issue is not only whether food supply is currently above a simple benchmark, but whether it is improving, remaining stagnant, or weakening under continued population growth. From this perspective, the lesson drawn from China is analytical rather than prescriptive: historical comparison may help identify patterns of pressure, but it cannot by itself provide a direct answer for countries operating under different structural conditions.

6 Conclusion

This thesis examines the relationship between population growth and national food availability in China, Egypt, and Nigeria through a comparative descriptive analysis based on secondary data from the World Bank and FAOSTAT. By focusing on two key indicators—population and food supply—it compares China’s historical experience with more recent developments in Egypt and Nigeria within a limited and clearly defined analytical framework.

As for Research Question 1, the results show that during 1961–1980 China experienced continuous and very large population growth together with relatively restricted food supply conditions. During this period, the population increased steadily, while food supply, although improving overall, remained low for most of the time and did not increase evenly. In most years, it stayed below the 2,100 kcal/person/day reference point used in this thesis, and only in 1980 did it move slightly above that level. Therefore, this period can reasonably be regarded as a stage of food-related pressure under rapid population growth.

Regarding RQ2, the results show that China followed a different pattern after 1980. Population growth continued, but at a much slower rate than before, while food supply increased steadily and remained above the benchmark throughout the period. The widening gap between food supply and the reference standard, together with the slowdown in population growth, indicates long-term improvement in national food availability. Therefore, this thesis treats the Chinese data after 1980 as supplementary evidence of a later stage in which food supply improved while population pressure became lower.

As for Research Question 3, the results for Egypt and Nigeria indicate sustained population growth during 2010–2023 without matching improvement in food supply. In Egypt, the population increased steadily while food supply declined overall, although it remained above the 2,100 kcal/person/day benchmark. In Nigeria, population growth was even stronger, and food supply ended lower than at the beginning of the period. In both cases, population expansion was not accompanied by stronger national food supply. Within the scope of this research, these trends may be interpreted as caution signals that food-related

pressure could become more difficult to manage if population continues to grow without corresponding improvement in food supply.

As regards RQ4, the comparative analysis provides only limited lessons. China's historical experience should not be treated as a direct example for Egypt or Nigeria, because the three cases differ in historical background, institutions, resources, and wider food-system conditions. However, it still shows that large population growth together with limited progress in food supply may be associated with lasting pressure on national food availability. From this perspective, China's earlier stage can serve as a useful historical reference point for understanding why recent trends in Egypt and Nigeria deserve attention.

This thesis makes a modest contribution by comparing the three countries within one simple and consistent analytical framework. First, for all three cases, it makes use of the same two indicators—population and food supply—thus making the comparison simpler and more direct. Second, it makes a clear distinction between the two phases in the Chinese case, instead of regarding China as a single unified experience. Third, this work makes examination of changes in the difference between food supply and the benchmark, not only whether supply lies above or under it, therefore, to find out the changing pressure on food availability across time. This lets it be possible to display not just low-level pressure, just like in China's earlier time, but also dropping food supply under lasting population growth, as is seen in Egypt and Nigeria.

At the same time, the conclusions of this thesis, therefore, must be comprehended within the scope of its existing limitations. This research depends on macro-level secondary data and does not carry out testing on causal connections. It likewise does not make an investigation into other important dimensions of food security—such as acquisition, distribution, management, conflict, water scarcity, import reliance, or policy change—with the same empirical depth. Due to this reason, this thesis does not put forward the claim that it can explain all causes of food insecurity or food production challenges in the three countries. However, within this limited comparative framework, it shows that the relationship between long-term population growth and weak food supply performance is worth attention.

Overall, within the scope and limits of this study, the results show that China during 1961–1980 can be understood as a historical phase of food-related pressure under rapid population growth. At the same time, recent developments in Egypt and Nigeria may be seen as caution signals that food production challenges could become harder to manage if population growth continues without corresponding improvement in national food supply.

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